LatticeSim

Hamiltonian lattice simulations of bosonized (1+1)-dimensional gauge theories

From quantum mechanics to the search for bound states

LatticeSim documentation, release 0.1

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Chapter 1

Introduction

1.1 What LatticeSim does

LatticeSim simulates quantum field theories in one space and one time dimension on a spatial lattice, in the Hamiltonian formulation. Its present scope is the class of theories that can be solved *exactly* on the lattice by linear algebra: *massless fermions coupled to abelian gauge fields*, $U(1)^k$, with arbitrary charges for the left- and right-moving fermions. After bosonization (Chapter 4) and elimination of the gauge field through the Gauss law (Chapter 5) such a theory becomes a *quadratic* Hamiltonian of coupled harmonic oscillators, and everything about it follows from one real symmetric matrix (Chapter 2).

The goal of this release is a complete, tested path from the definition of a theory to the question

> *Which gauge-invariant particles does the theory contain, and which of them are bound states of the elementary fermions?*

The answer for the 3-4-5-0 chiral gauge theory, a standard test case, is worked out in Chapter 7: the theory contains a massless gauge-invariant composite fermion $\psi_{3L}^2 \psi_{4L} \psi_{5R}^{-2}$ and its antiparticle, found both from the lattice Hamiltonian and from an independent analytic argument.

1.2 How to read this document

The prerequisites are quantum mechanics (harmonic oscillator, commutators, Heisenberg picture) and linear algebra. Everything else is developed step by step:

Chapter 2
Quadratic Hamiltonians, normal modes, the Williamson normal form, Gaussian ground states and zero modes. This is the numerical core of the library.

Chapter 3
A free scalar field on a lattice: discretization, boundary conditions, the continuum limit, and the zero modes of a field on a ring.

Chapter 4
Fermions in one dimension and their description by a compact boson; why the fermion numbers are windings and momenta of the boson.

Chapter 5
Gauge fields in one dimension: the Gauss law, the Schwinger model and its mass, correlation functions, screening and confinement of probe charges, the θ angle.

Chapter 6
Gauge anomalies, the anomaly-cancellation condition, and which global symmetries survive the gauging.

Chapter 7
Charge sectors, the bound-state search, and the 3-4-5-0 model as worked example, including a recipe for testing your own theory.

Each chapter states which part of the library implements it, and the figures are produced by the studies that come with the library, so every number in this document can be reproduced.

1.3 Quick start

Installation (Python ≥ 3.10) into a virtual environment, from the repository root:

```
python3 -m venv .venv
source .venv/bin/activate
pip install -e ".[dev]"
pytest # fast tests; "pytest -m slow" runs the expensive ones
```

The bound-state search of Chapter 7 in a few lines:

```
from latsimpkg import ChiralTheory, find_bound_states

theory = ChiralTheory(QL=[3, 4], QR=[5, 0], g=0.3) # charges of left and right
    movers
scan = find_bound_states(theory)

print(scan.agrees) # lattice and analytic prediction agree
for s in scan.bound_states:
    print(s.content, s.h, s.hb) # psi_3L^2 psi_4L psi_5R^-2 0.5 0.0
```

Studies, figures and this document:

```
python -m studies # run all studies, results -> results/
python -m studies.figures # figures -> figures/ (only what is out of date)
python tools/build_docs.py # this PDF -> docs/build/latticesim.pdf
```

1.4 Structure of the library

The package `latsimpkg` is built in layers; every layer only uses the layers above it in this list.

module	content
`conventions`	normalizations and sign conventions (Appendix A)
`theory`	`ChiralTheory`: charges, anomaly check, pairing, meson masses
`lattice`	`Lattice`: number of sites, spacing, open chain or ring
`phasespace`	`PhaseSpace`: layout of the vector of coordinates and momenta
`quadratic`	`QuadraticHamiltonian`: normal modes, Williamson form, ground state, zero modes
`models`	`build_gauge_model`: theory + lattice $\rightarrow$ Hamiltonian
`analysis`	spectra, correlators, potentials, charge sectors, result types
`api`	`find_bound_states`, `spectrum`, `screening_potential`
`plotting`	figures of saved results

The API reference (Appendix B) is generated from the docstrings of these modules.

Chapter 2

Quadratic Hamiltonians and Gaussian states

Every model in this library reduces to a system of coupled harmonic oscillators. This chapter develops the complete theory of such systems: equations of motion, normal modes, the ground state, and the special role of zero modes. It is implemented in `latsimpkg.quadratic.QuadraticHamiltonian`.

2.1 Phase space and the symplectic matrix

Consider n degrees of freedom with coordinates $q_1, \dots, q_n$ and momenta $p_1, \dots, p_n$, collected in the *phase-space vector*

$$\boldsymbol{X} = (q_1, \dots, q_n, p_1, \dots, p_n)^T. \tag{2.1}$$

The canonical commutation relations $[q_j, p_k] = i\delta_{jk}$, $[q_j, q_k] = [p_j, p_k] = 0$ ($\hbar = 1$) take the compact form

$$[X_a, X_b] = iJ_{ab}, \qquad J = \begin{pmatrix} 0 & \mathbb{1}_n \\ -\mathbb{1}_n & 0 \end{pmatrix}. \tag{2.2}$$

J is the *symplectic matrix*; it is real, antisymmetric, orthogonal, and $J^2 = -\mathbb{1}$. In the library the layout of $\boldsymbol{X}$ is described by a `PhaseSpace` with named blocks (for a field: one block of coordinates $\varphi_{i,j}$ per flavor i and site j, and its conjugate momenta), and `PhaseSpace.J` is the matrix (2.2).

2.2 Quadratic Hamiltonians and equations of motion

A quadratic Hamiltonian is

$$H = \tfrac{1}{2}\boldsymbol{X}^T M \boldsymbol{X} = \tfrac{1}{2}\sum_{a,b} M_{ab} X_a X_b, \tag{2.3}$$

with a real symmetric $2n \times 2n$ matrix M. Stability requires M to be positive semidefinite, $\boldsymbol{X}^T M \boldsymbol{X} \geq 0$, otherwise the energy is unbounded from below.

The Heisenberg equation $\dot{X}_a = i[H, X_a]$ follows from (2.2) and $[X_b X_c, X_a] = X_b[X_c, X_a] + [X_b, X_a]X_c$:

$$[H, X_a] = \tfrac{1}{2} M_{bc}(iX_b J_{ca} + iJ_{ba} X_c) = -i(JM\boldsymbol{X})_a, \tag{2.4}$$

using $J^T = -J$ and the symmetry of M. Hence

$$\dot{\boldsymbol{X}} = JM\boldsymbol{X}, \qquad \boldsymbol{X}(t) = e^{JMt}\boldsymbol{X}(0). \tag{2.5}$$

The operator equations are linear and identical to the classical Hamilton equations; everything about the dynamics is encoded in the matrix JM.

Example 2.1 (single oscillator)**.** For $H = \frac{1}{2}(\omega^2 q^2 + p^2)$ we have $M = \mathrm{diag}(\omega^2, 1)$ and $JM = \begin{pmatrix} 0 & 1 \\ -\omega^2 & 0 \end{pmatrix}$ with eigenvalues $\pm i\omega$.

2.3 Normal modes and the Williamson normal form

A linear change of variables $\boldsymbol{X} = S\boldsymbol{Y}$ preserves the commutation relations, $[Y_a,\, Y_b] = iJ_{ab}$, if and only if

$$SJS^T = J, \tag{2.6}$$

i.e. if S is *symplectic*. The quantum mechanics of quadratic Hamiltonians is therefore the linear algebra of symplectic transformations. The central result is

Theorem 2.1 (Williamson [1])**.** *For every real symmetric positive definite M there is a symplectic S with*

$$S^T M S = \begin{pmatrix} \Omega & 0 \\ 0 & \Omega \end{pmatrix}, \qquad \Omega = \mathrm{diag}(\omega_1, \dots, \omega_n),\ \omega_k > 0. \tag{2.7}$$

In the new variables $\boldsymbol{Y} = (Q_1, \dots, Q_n, P_1, \dots, P_n)$ the Hamiltonian decouples into independent oscillators,

$$H = \sum_{k=1}^{n} \frac{\omega_k}{2}(Q_k^2 + P_k^2) = \sum_k \omega_k (a_k^\dagger a_k + \tfrac{1}{2}), \qquad a_k = \tfrac{1}{\sqrt{2}}(Q_k + iP_k). \tag{2.8}$$

The ω_k are the *normal-mode frequencies*; the eigenvalues of JM are $\pm i\omega_k$. The ground state $|0\rangle$ is annihilated by all a_k and has the *zero-point energy*

$$E_0 = \frac{1}{2}\sum_k \omega_k. \tag{2.9}$$

Excited states are obtained by acting with $a_k^\dagger$; in field theory each $a_k^\dagger$ creates a particle of energy ω_k.

How the library computes it. Diagonalizing the non-symmetric matrix JM is numerically delicate. Instead, $JM = M^{-1/2}\,(M^{1/2} J M^{1/2})\,M^{1/2}$ is similar to the antisymmetric matrix $A = M^{1/2} J M^{1/2}$, so the frequencies are the positive eigenvalues of the *Hermitian* matrix iA. Its eigenvectors $v = (x + iy)/\sqrt{2}$ for eigenvalue ω satisfy $Ax = \omega y$, $Ay = -\omega x$. With the orthogonal $O = (y_1, \dots, y_n, x_1, \dots, x_n)$ one has $O^T A O = \begin{pmatrix} 0 & \Omega \\ -\Omega & 0 \end{pmatrix}$, and

$$S = M^{-1/2}\, O \begin{pmatrix} \Omega^{1/2} & 0 \\ 0 & \Omega^{1/2} \end{pmatrix} \tag{2.10}$$

is the Williamson transformation: $S^T M S = \mathrm{diag}(\Omega, \Omega)$ and $SJS^T = J$ (check it using $M^{-1/2} J M^{-1/2} = -A^{-1}$). This is `QuadraticHamiltonian.frequencies` and `.williamson`.

2.4 The ground state: Gaussian states and the covariance matrix

The ground state of a quadratic Hamiltonian is a *Gaussian state*: all its properties follow from the first and second moments. By symmetry $\langle \boldsymbol{X} \rangle = 0$, and the second moments are collected in the *covariance matrix*

$$\sigma_{ab} = \tfrac{1}{2}\langle \{X_a,\, X_b\} \rangle. \tag{2.11}$$

In normal-mode variables the oscillator ground state has $\langle Q_k^2 \rangle = \langle P_k^2 \rangle = \tfrac{1}{2}$ and no correlations, so $\sigma_Y = \frac{1}{2}\mathbb{1}$, and transforming back,

$$\sigma = \tfrac{1}{2} S S^T = \tfrac{1}{2} M^{-1/2} \left| M^{1/2} J M^{1/2} \right| M^{-1/2}, \tag{2.12}$$

where $|A| = \sqrt{A^T A}$. For a single oscillator this gives the familiar $\langle q^2 \rangle = 1/(2\omega)$ and $\langle p^2 \rangle = \omega/2$. Any correlation function of linear operators follows, e.g. $\langle (\boldsymbol{u}^T \boldsymbol{X})(\boldsymbol{v}^T \boldsymbol{X}) \rangle_{\mathrm{sym}} = \boldsymbol{u}^T \sigma \boldsymbol{v}$; the electric-field correlator of Chapter 5 is computed this way.

Uncertainty principle and purity. For every quantum state

$$\sigma + \tfrac{i}{2} J \;\geq\; 0 \tag{2.13}$$

(as a Hermitian matrix), the matrix form of the Heisenberg uncertainty relation [2]. A Gaussian state is *pure* exactly if the smallest eigenvalue of $\sigma + \frac{i}{2}J$ is zero: all Williamson eigenvalues of σ equal $\tfrac{1}{2}$. The tests check this for every ground state the library produces.

2.5 Zero modes: free particles and conserved charges

If M has a kernel, Theorem 2.1 does not apply. The simplest example is a free particle, $H = p^2/2m$: $M = \mathrm{diag}(0, 1/m)$, the coordinate q is a *flat direction*, and the momentum p is conserved. On a ring the uniform shift of a field is such a flat direction, and its conserved momentum turns out to be a fermion number (Chapter 4); zero modes are therefore not a nuisance but the place where the charges live.

Conserved linear charges. A linear operator $\boldsymbol{w}^T \boldsymbol{X}$ is conserved if $\frac{d}{dt} \boldsymbol{w}^T \boldsymbol{X} = \boldsymbol{w}^T J M \boldsymbol{X} = 0$ for all $\boldsymbol{X}$, i.e.

$$M J \boldsymbol{w} = 0 \quad \Longleftrightarrow \quad \boldsymbol{w} \in J \ker M. \tag{2.14}$$

With an orthonormal basis K of $\ker M$ the conserved charges are the columns of $A = JK$.

Normal form with zero modes. We require the kernel to be *isotropic*, $K^T J K = 0$: no two flat directions are conjugate to each other (a pair q, p with H independent of both would have no dynamics at all and no ground state). Then every flat direction is a free particle, and the phase space splits into

- the flat directions K (positions of the free particles),
- their conjugates, measured by the conserved charges $\boldsymbol{c} = A^T \boldsymbol{X}$ (momenta of the free particles),
- the oscillator space $W = \{ \boldsymbol{X} : K^T \boldsymbol{X} = 0,\; K^T J \boldsymbol{X} = 0 \}$.

On W the matrix M is positive definite and the Williamson construction applies with the symplectic form $\Omega_W = B^T J B$ of a basis B of W. Every $\boldsymbol{X}$ decomposes uniquely as $\boldsymbol{X} = M^+ A G \boldsymbol{c} + B\boldsymbol{\xi} + K\boldsymbol{\kappa}$ with the pseudo-inverse M^+ and $G = (A^T M^+ A)^{-1}$, and the energy separates exactly:

$$H = \tfrac{1}{2}\,\boldsymbol{c}^T G\,\boldsymbol{c} \;+\; \tfrac{1}{2}\,\boldsymbol{\xi}^T (B^T M B)\,\boldsymbol{\xi}. \tag{2.15}$$

The first term is the kinetic energy of the free particles, the second a set of oscillators.

Lowest energy at fixed charges. Because the two terms in (2.15) are independent, the lowest energy in the sector with conserved charges $\boldsymbol{c}$ is $\frac{1}{2}\boldsymbol{c}^T G \boldsymbol{c}$ above the oscillator ground state. It equals the classical constrained minimum, which follows from a Lagrange multiplier $\boldsymbol{\lambda}$:

$$\min_{A^T \boldsymbol{X} = \boldsymbol{c}} \tfrac{1}{2}\boldsymbol{X}^T M \boldsymbol{X} : \quad M\boldsymbol{X} = A\boldsymbol{\lambda} \;\Rightarrow\; \boldsymbol{X}^* = M^+ A G \boldsymbol{c}, \quad E(\boldsymbol{c}) = \tfrac{1}{2}\,\boldsymbol{c}^T G\,\boldsymbol{c}. \tag{2.16}$$

This formula is the engine of the bound-state search (Chapter 7). In the library it is implemented by the methods `free_energy` and `constrained_minimum` of `QuadraticHamiltonian`. The ground-state covariance with zero modes, `.covariance`, describes the oscillators at sharp conserved charges; the flat directions themselves have infinite variance.

2.6 In code

```
import numpy as np
from latsimpkg.quadratic import QuadraticHamiltonian

H = QuadraticHamiltonian(np.diag([1.7**2, 1.0])) # one oscillator, omega = 1.7
H.frequencies # [1.7]
H.covariance[0, 0] # <q^2> = 1/(2 omega)
H.ground_state_energy # 0.85

free = QuadraticHamiltonian(np.diag([0.0, 1.0])) # free particle H = p^2/2
free.n_free, free.conserved_charges # 1, A = J K: the charge c = -p
free.free_energy([2.0]) # c = 2 (p = -2) -> E = 2
```

`QuadraticHamiltonian` checks its input: M must be symmetric, positive semidefinite, finite, match the phase space and have an isotropic kernel; otherwise it raises an error instead of returning wrong numbers. The test suite verifies the construction on random symplectic matrices with known frequencies and free particles.

Chapter 3

A free scalar field on a lattice

A quantum field in one spatial dimension is an infinite collection of coupled oscillators, one per point x. Replacing the continuum by a lattice of N sites turns it into exactly the finite system of Chapter 2; this Hamiltonian lattice formulation goes back to Kogut and Susskind [8]. This chapter introduces the discretization used throughout the library, the two boundary conditions, and the continuum limit.

3.1 The continuum theory

A real scalar field $\varphi(x)$ with conjugate momentum $\Pi(x)$, $[\varphi(x),\,\Pi(y)] = i\delta(x-y)$, has the Hamiltonian

$$H = \int dx \left[\tfrac{1}{2}\Pi^2 + \tfrac{1}{2}(\partial_x\varphi)^2 + \tfrac{1}{2}\mu^2\varphi^2\right]. \tag{3.1}$$

Plane waves e^{ikx} are normal modes with $\omega_k = \sqrt{\mu^2 + k^2}$: the field describes free particles of mass μ (speed of light $c = 1$). For $\mu = 0$ the particles are massless and move at the speed of light.

3.2 Discretization

Put the field on sites $x_j = ja$, $j = 0, \ldots, N-1$, with lattice spacing a, and define

$$\varphi_j = \varphi(x_j), \qquad P_j = a\,\Pi(x_j). \tag{3.2}$$

The factor a makes the lattice variables canonical: the continuum $\delta(x-y)$ becomes δ_{jk}/a, so $[\varphi_j,\,P_k] = i\delta_{jk}$. Replacing $\int dx \to a\sum_j$ and $\partial_x\varphi \to (\varphi_{j+1}-\varphi_j)/a$ gives

$$H = \sum_j \Big[\frac{P_j^2}{2a} + \frac{(\varphi_{j+1}-\varphi_j)^2}{2a}\Big] = \tfrac{1}{2}\boldsymbol{X}^T M \boldsymbol{X}, \qquad \boldsymbol{X} = (\varphi_0, \ldots, \varphi_{N-1}, P_0, \ldots, P_{N-1}). \tag{3.3}$$

The differences $\ell_j = \varphi_{j+1} - \varphi_j$ live on the *links* between sites; the library represents them by a matrix B with $\ell = B\boldsymbol{X}$, so that the gradient energy is $\boldsymbol{X}^T B^T B \boldsymbol{X}/(2a)$.

3.3 Open chain with Dirichlet boundary conditions

On the open chain the field is fixed to zero just outside the lattice, $\varphi_{-1} = \varphi_N = 0$. There are $N+1$ links and

$$M = \frac{1}{a}\begin{pmatrix} K & 0 \\ 0 & \mathbb{1} \end{pmatrix}, \qquad K = \begin{pmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & \ddots & \ddots & \ddots \\ & & -1 & 2 \end{pmatrix}. \tag{3.4}$$

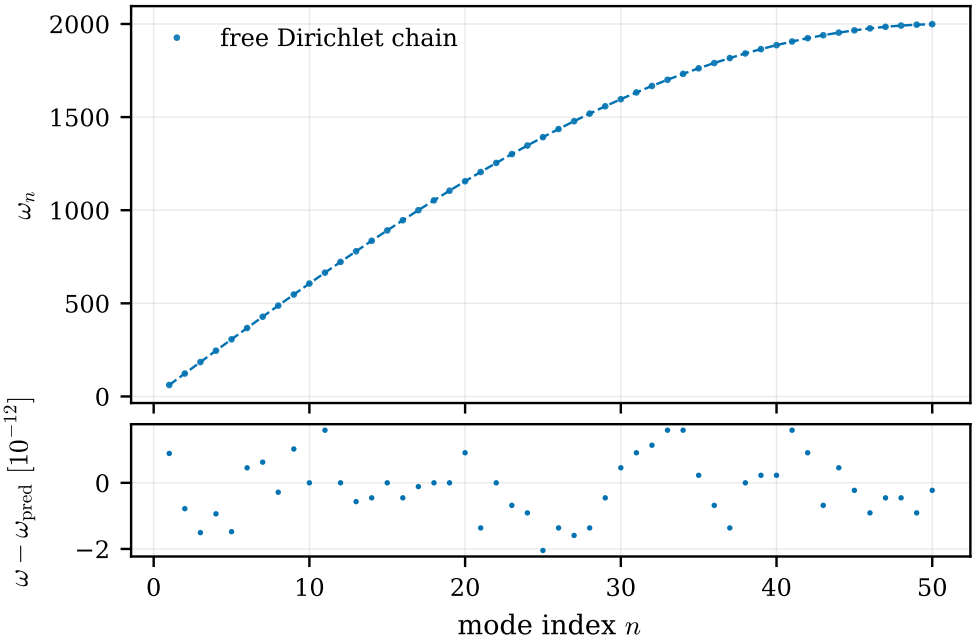


Figure 3.1: Normal-mode frequencies of the free scalar field on the Dirichlet chain ($N = 50$, $a = 0.001$) compared with $\omega_n = (2/a)\sin(\pi n/2(N+1))$ (dashed). The residuals are at round-off level.

The eigenvectors of K are standing waves $\sin(k_n x_j)$ with

$$k_n = \frac{\pi n}{L}, \quad L = (N+1)a, \qquad \omega_n = \frac{2}{a}\sin\frac{k_n a}{2}, \qquad n = 1, \dots, N. \tag{3.5}$$

L is the distance between the two fixed points. For $k_n a \ll 1$ the lattice dispersion reproduces the continuum, $\omega_n = k_n(1 - (k_n a)^2/24 + \dots)$: lattice artefacts are of order $(ka)^2$. Figure 3.1 compares the numerical frequencies with (3.5).

Ultraviolet behaviour. The zero-point energy $E_0 = \frac{1}{2}\sum_n \omega_n$ grows like N/a: it is dominated by the shortest wavelengths and has no continuum limit. Physical statements always concern *differences* of energies (masses, excitation energies, potentials), which do have a limit.

3.4 The ring and its zero modes

On a ring (periodic boundary conditions, $\varphi_N \equiv \varphi_0$, $L = Na$) the momenta are $k_n = 2\pi n/L$, and the mode $n = 0$ is special: shifting the whole field, $\varphi_j \to \varphi_j + c$, costs no energy. In the language of Section 2.5 the uniform shift is a flat direction, and the total momentum

$$P_{\mathrm{tot}} = \sum_j P_j \tag{3.6}$$

is a conserved charge with kinetic energy $P_{\mathrm{tot}}^2/(2L)$ (each $P_j = P_{\mathrm{tot}}/N$ contributes $P_j^2/2a$).

Compact fields and windings. In the theories of this library the field is *compact*: φ and $\varphi + \sqrt{\pi}$ describe the same physical configuration (Chapter 4 explains why). Then the field may wind around the ring,

$$\varphi(x + L) = \varphi(x) + \sqrt{\pi}\, n, \qquad n \in \mathbb{Z}, \tag{3.7}$$

and the winding number n cannot change under continuous time evolution. The library writes such a field as a periodic part plus a uniform slope, $\varphi_j = \tilde{\varphi}_j + Wj/N$ with $W = \sqrt{\pi}n$. Because the periodic differences sum to zero, the gradient energy splits into $\sum_j (\tilde{\varphi}_{j+1} - \tilde{\varphi}_j)^2/2a + W^2/(2L)$. W is treated as a canonical momentum, conjugate to a new coordinate $\bar{\vartheta}$ (the *dual zero mode*, Chapter 4): the phase space on the ring is

$$\boldsymbol{X} = (\tilde{\varphi}_0, \ldots, \tilde{\varphi}_{N-1},\ \bar{\vartheta},\ P_0, \ldots, P_{N-1},\ W), \tag{3.8}$$

of dimension $2(N+1)$ per field. Both P_{tot} and W are conserved for the free field, and the energy of a configuration with given values is

$$E = \frac{P_{\text{tot}}^2}{2L} + \frac{W^2}{2L}. \tag{3.9}$$

With $P_{\text{tot}} = \sqrt{\pi}m$ and $W = \sqrt{\pi}n$ this is $E = \pi(m^2 + n^2)/(2L)$: an energy proportional to $1/L$, the hallmark of a massless particle in a box. Chapter 4 identifies n and m with fermion numbers.

3.5 In code

```
from latsimpkg import ChiralTheory, Lattice, build_gauge_model

free = ChiralTheory([0], [0], g=0.0) # one neutral fermion = one free boson
open_chain = build_gauge_model(free, Lattice(50, a=0.1, boundary="open"))
open_chain.H.frequencies # (2/a) sin(k_n a/2)

ring = build_gauge_model(free, Lattice(20, boundary="ring"))
ring.H.n_free # 2 conserved charges: P_tot and W
```

Chapter 4

Fermions in one dimension and bosonization

The theories of interest contain fermions, but the library simulates bosons. In one spatial dimension this is not an approximation: a massless Dirac fermion *is* a compact massless boson, a statement known as bosonization [6, 7, 9]. This chapter explains the part of the dictionary the library uses and checks it on the spectrum of free fermions on a ring.

4.1 Chiral fermions

In one dimension a massless fermion field splits into a left-moving and a right-moving component, $\psi = (\psi_L, \psi_R)$, with

$$H = \int dx \left[\psi_R^\dagger(-i\partial_x)\psi_R + \psi_L^\dagger(i\partial_x)\psi_L\right]. \tag{4.1}$$

A right mover has energy $E = k$ for momentum k, a left mover $E = -k$; both move at the speed of light and never scatter into each other without interactions. Each component has its own particle number, $N_L = \int \psi_L^\dagger\psi_L$ and $N_R = \int \psi_R^\dagger\psi_R$ (measured relative to the filled Dirac sea). A *chiral* theory treats them differently, for example by giving them different gauge charges.

Spectrum on a ring. On a ring of length L (with antiperiodic boundary conditions for the fermions) the single-particle momenta are $(r + \frac{1}{2})\,2\pi/L$. Adding N_R right movers above the Fermi sea fills the lowest levels and costs

$$\frac{2\pi}{L}\sum_{r=0}^{N_R-1}\left(r+\tfrac{1}{2}\right) = \frac{2\pi}{L}\,\frac{N_R^2}{2}, \tag{4.2}$$

the same for holes ($N_R < 0$) and for left movers. The lowest energy at given fermion numbers is therefore

$$E(N_L, N_R) = \frac{2\pi}{L}\,(h + \bar{h}), \qquad h = \frac{N_L^2}{2}, \quad \bar{h} = \frac{N_R^2}{2}, \tag{4.3}$$

and its momentum is $P = \frac{2\pi}{L}(\bar{h} - h)$ (left movers carry negative momentum). Above these states there are particle-hole excitations with energies in multiples of $2\pi/L$. The numbers $h, \bar{h}$ are the *conformal weights* of the state; $\Delta = h + \bar{h}$ is its scaling dimension and $s = h - \bar{h}$ its spin. A single fermion has $\Delta = |s| = \frac{1}{2}$.

4.2 The bosonization dictionary

Introduce a scalar field φ with momentum Π such that the vector (total) and axial (left minus right) fermion densities are

$$\rho_V = \psi_L^\dagger \psi_L + \psi_R^\dagger \psi_R = \frac{1}{\sqrt{\pi}}\,\partial_x \varphi, \qquad \rho_A = \psi_L^\dagger \psi_L - \psi_R^\dagger \psi_R = \frac{1}{\sqrt{\pi}}\,\Pi. \tag{4.4}$$

The Hamiltonian of the free fermion becomes the free massless boson, $H = \int \frac{1}{2}[\Pi^2 + (\partial_x \varphi)^2]$, and the particle-hole excitations of the fermions are the oscillator modes of the boson. It is convenient to define the *dual field*

$$\vartheta(x) = \int^x dy\, \Pi(y), \qquad \rho_A = \frac{1}{\sqrt{\pi}} \partial_x \vartheta, \tag{4.5}$$

so that φ counts vector charge and ϑ axial charge in the same way.

Fermion numbers are zero modes. Integrating (4.4) over the ring,

$$n \equiv N_L + N_R = \frac{1}{\sqrt{\pi}}[\varphi(L) - \varphi(0)], \qquad m \equiv N_L - N_R = \frac{1}{\sqrt{\pi}} \int_0^L \Pi. \tag{4.6}$$

The vector fermion number is the *winding* of φ and the axial number its *total momentum*. Since fermion numbers are integers, φ must be compact with period $\sqrt{\pi}$ (Section 3.4), and n and m are integers with

$$n \equiv m \pmod 2, \tag{4.7}$$

because $N_L = (n+m)/2$ and $N_R = (n-m)/2$ are integers. This parity rule is what makes the compact boson fermionic: $(n, m) = (1, 0)$ would be half a left and half a right fermion.

Check. The zero-mode energy of the boson (3.9) with $W = \sqrt{\pi} n$, $P_{\text{tot}} = \sqrt{\pi} m$ is

$$E = \frac{\pi(n^2 + m^2)}{2L} = \frac{\pi}{L}(N_L^2 + N_R^2) = \frac{2\pi}{L}\Big(\frac{N_L^2}{2} + \frac{N_R^2}{2}\Big), \tag{4.8}$$

exactly the fermionic result (4.3). The lattice momentum $P = -\frac{1}{a}\sum_j P_j \ell_j$ gives

$$s = -\frac{PL}{2\pi} = \frac{nm}{2} = \frac{N_L^2 - N_R^2}{2} = h - \bar{h}. \tag{4.9}$$

The library reproduces both exactly on any ring (`analysis.sectors.sector_energy`; tested for several (N_L, N_R)). Note the sign convention: left movers have negative momentum and positive spin, $s = h - \bar{h}$.

4.3 Many fermions and their charges

A theory contains left movers $L_1, \ldots, L_{n_L}$ and right movers $R_1, \ldots, R_{n_R}$, with charges $Q_{L,i}$, $Q_{R,j}$ under a $U(1)$ gauge field (for $U(1)^k$: one row of charges per gauge field). To bosonize, the library pairs L_i with R_i into a Dirac fermion and describes each pair by one compact boson φ_i. The gauge current $\sum_i (Q_{L,i}\rho_{L,i} + Q_{R,i}\rho_{R,i})$ becomes

$$\sum_i (Q_i^V \rho_{V,i} + Q_i^A \rho_{A,i}), \qquad Q^V = \frac{Q_L + Q_R}{2}, \quad Q^A = \frac{Q_L - Q_R}{2}. \tag{4.10}$$

The pairing is bookkeeping: physical results must not depend on it, and the tests check this explicitly for the two possible pairings of the 3-4-5-0 model. If $n_L \neq n_R$, the missing partners are added as neutral fermions (charge 0), which are free and decoupled (`ChiralTheory.pairing`).

```python
from latsimpkg import ChiralTheory

t = ChiralTheory(QL=[3, 4], QR=[5, 0], g=0.3)
p = t.pairing()
p.QV, p.QA # [[4, 2]], [[-1, 2]]
t.pairing(order=[1, 0]).QV # the other pairing: [[1.5, 4.5]]
```

Chapter 5

Gauge fields in one dimension

In one spatial dimension the electromagnetic field has no transverse polarizations: there are no photons. The electric field is completely fixed by the charges through the Gauss law, and eliminating it leaves an interaction between the charges. For bosonized fermions this interaction is quadratic, so the gauge theory stays a system of coupled oscillators. This chapter derives the lattice Hamiltonian used by `models.build_gauge_model` and its main physical consequences.

5.1 Electric field and Gauss law

In the temporal gauge $A_0 = 0$ the only field is $A_1(x)$ with conjugate momentum $-E(x)$, and the Hamiltonian of the gauge field is $\frac{1}{2}\int E^2$ (there is no magnetic field in one dimension). Physical states satisfy the Gauss law

$$\partial_x E = g\,\rho, \tag{5.1}$$

where ρ is the charge density of the matter and g the coupling. With the bosonized densities (4.4) and charges (4.10),

$$g\rho = \frac{g}{\sqrt{\pi}}\sum_i (Q_i^V\,\partial_x\varphi_i + Q_i^A\,\partial_x\vartheta_i) \quad\Rightarrow\quad E(x) = \frac{g}{\sqrt{\pi}}\sum_i (Q_i^V\,\varphi_i(x) + Q_i^A\vartheta_i(x)) + e(x). \tag{5.2}$$

The integration constant $e(x)$ is a c-number background: a constant $e = g\theta/2\pi$ (the θ angle) or the field of static external charges (Section 5.5). The total Hamiltonian is

$$H = \int dx\,\Big[\tfrac{1}{2}\sum_i (\Pi_i^2 + (\partial_x\varphi_i)^2) + \tfrac{1}{2}E^2\Big]. \tag{5.3}$$

With several gauge fields $U(1)^k$ there is one E_a with coupling g_a and charges Q_a^V, Q_a^A per gauge field, and the electric energy is $\frac{1}{2}\sum_a \int E_a^2$.

On the lattice. With $\vartheta_{i,j} = \sum_{k\le j} P_{i,k}$ (the discrete version of $\int^x \Pi$) the electric field on site j is a linear function of $\boldsymbol{X}$,

$$E_j = (C\boldsymbol{X})_j + e_j, \qquad E_j = \frac{g}{\sqrt{\pi}}\sum_i \Big(Q_i^V\,\varphi_{i,j} + Q_i^A\sum_{k\le j} P_{i,k}\Big) + e_j, \tag{5.4}$$

and the electric energy $\frac{a}{2}\sum_j E_j^2$ adds aC^TC to the matrix M (plus a linear term and a constant if $e \neq 0$). On the ring the zero modes enter as well: $\varphi_{i,j} = \tilde{\varphi}_{i,j} + W_i j/N$ and $\vartheta_{i,j} = \bar{\vartheta}_i + \sum_{k\le j} P_{i,k}$.

5.2 The Schwinger model

The Schwinger model [3, 4] is a single Dirac fermion with charge 1 ($Q_L = Q_R = 1$, so $Q^V = 1$, $Q^A = 0$). Then $E = (g/\sqrt{\pi})\varphi$ and (5.3) becomes

$$H = \int dx \left[\tfrac{1}{2}\Pi^2 + \tfrac{1}{2}(\partial_x\varphi)^2 + \frac{g^2}{2\pi}\varphi^2\right] : \tag{5.5}$$

a free boson of mass

$$\mu = \frac{g}{\sqrt{\pi}}. \tag{5.6}$$

The massless fermions have disappeared from the spectrum; the only particle is a massive neutral boson, the "meson". Its mass is generated non-perturbatively by the gauge field.

General charges. For one gauge field with charges Q_L, Q_R (anomaly free, Chapter 6) exactly one combination of the bosons becomes massive, with

$$\mu^2 = \frac{g^2}{\pi}\sum_i [(Q_i^V)^2 + (Q_i^A)^2] = \frac{g^2}{2\pi}\sum (Q_L^2 + Q_R^2), \tag{5.7}$$

and the remaining $F - 1$ combinations stay massless. For $U(1)^k$ the masses are the square roots of the eigenvalues of $\mu^2_{ab} = (g_a g_b/2\pi)(Q_{L,a}\cdot Q_{L,b} + Q_{R,a}\cdot Q_{R,b})$ (`ChiralTheory.meson_masses`).

5.3 Spectra on the open chain

On the open chain the predicted spectrum is one branch $\sqrt{\mu^2 + k_n^2}$ and $F - 1$ massless branches k_n, with the Dirichlet momenta (3.5). For vector charges the electric field acts locally on φ and the prediction is exact on the lattice. Axial charges act on ϑ, a sum over all sites to the left, whose boundary conditions differ from Dirichlet: the momenta become $k_n = (n + \delta)\pi/L$ with a phase shift δ, and the prediction holds only in the bulk. Figures 5.1 and 5.2 show both effects; the deviation for axial charges falls like $1/N$.

Continuum limit. At fixed physical length L the lowest frequency of the Schwinger model approaches $\sqrt{\mu^2 + (\pi/L)^2}$ as $a \to 0$, with the $(ka)^2$ lattice correction of Chapter 3 (Fig. 5.3).

5.4 Measuring the mass: the electric-field correlator

The electric field is proportional to the massive boson, so its correlations measure μ. For a free boson of mass μ the equal-time correlator in the ground state is

$$\langle \Phi(x)\Phi(0)\rangle = \int \frac{dk}{2\pi}\,\frac{e^{ikx}}{2\sqrt{k^2 + \mu^2}} = \frac{1}{2\pi}K_0(\mu|x|), \tag{5.8}$$

with the modified Bessel function K_0, which decays like $e^{-\mu|x|}$. The library computes $\langle E_j E_k\rangle = (C\sigma C^T)_{jk}$ from the covariance matrix (2.12) and fits $AK_0(\mu r) + B$ in a window away from the boundaries (`analysis.correlators.meson_mass_from_E` with an explicit `FitWindow`). The offset B is a uniform fluctuation of the background field caused by the open boundary, non-zero for axial charges. The fitted masses agree with (5.7) to about 10^{-3} (Fig. 5.4).

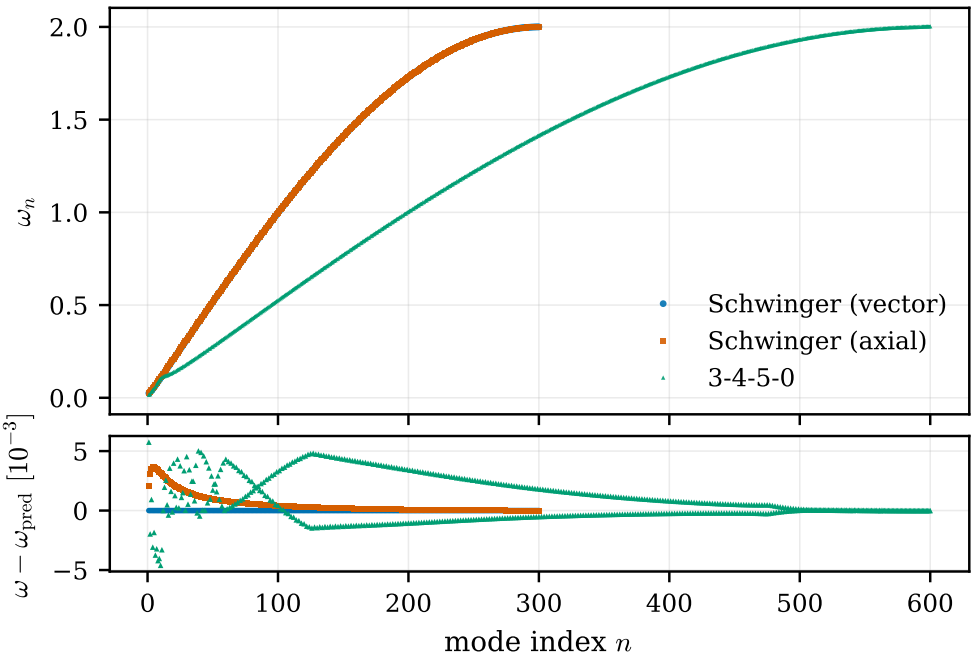


Figure 5.1: Spectra of the gauged theories on the open chain ($N = 300$, $g = 0.04$). Dashed: one branch $\sqrt{\mu^2 + k^2}$ plus $F - 1$ massless branches. Vector couplings agree exactly; axial couplings shift the boundary momenta, visible as residuals at small n.

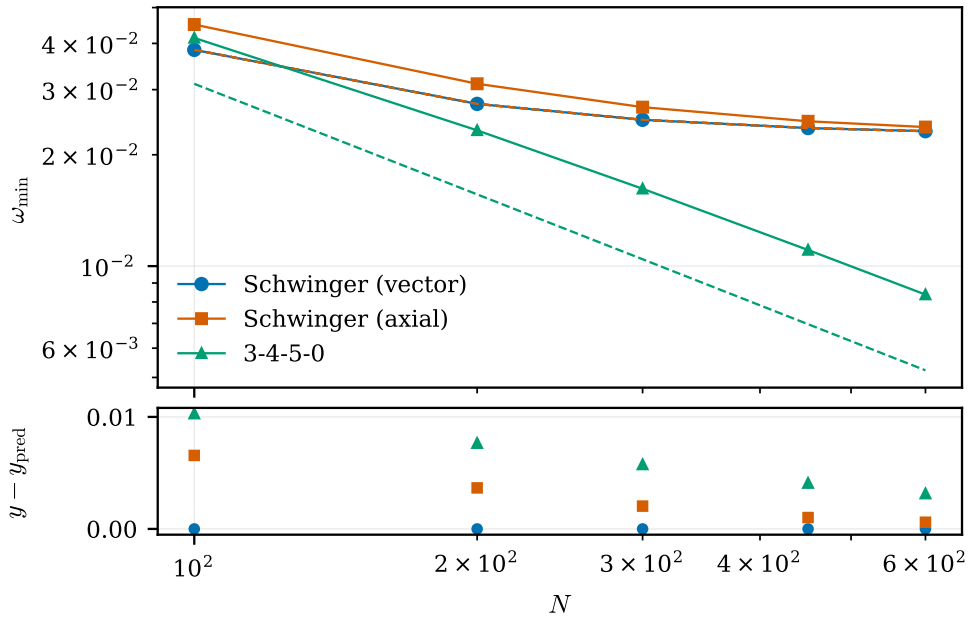


Figure 5.2: Lowest frequency against the chain length. With only a massive branch it approaches the meson mass μ; with massless flavors (3-4-5-0) it falls like $1/N$. Dashed: Dirichlet prediction.

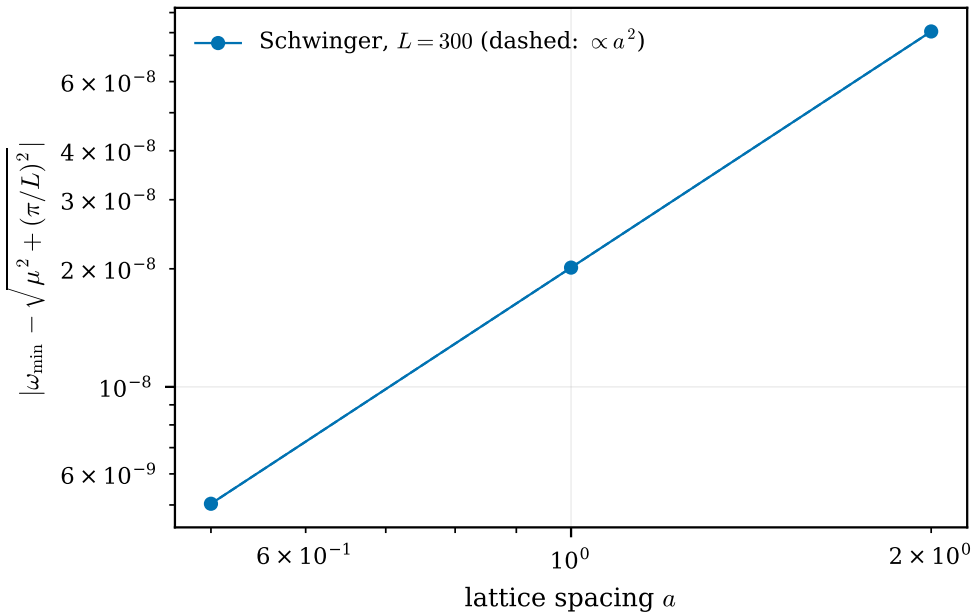


Figure 5.3: Continuum limit at fixed length $L = (N + 1)a = 300$ (Schwinger model, $g = 0.04$): the deviation from $\sqrt{\mu^2 + (\pi/L)^2}$ falls like a^2 (dashed reference line).

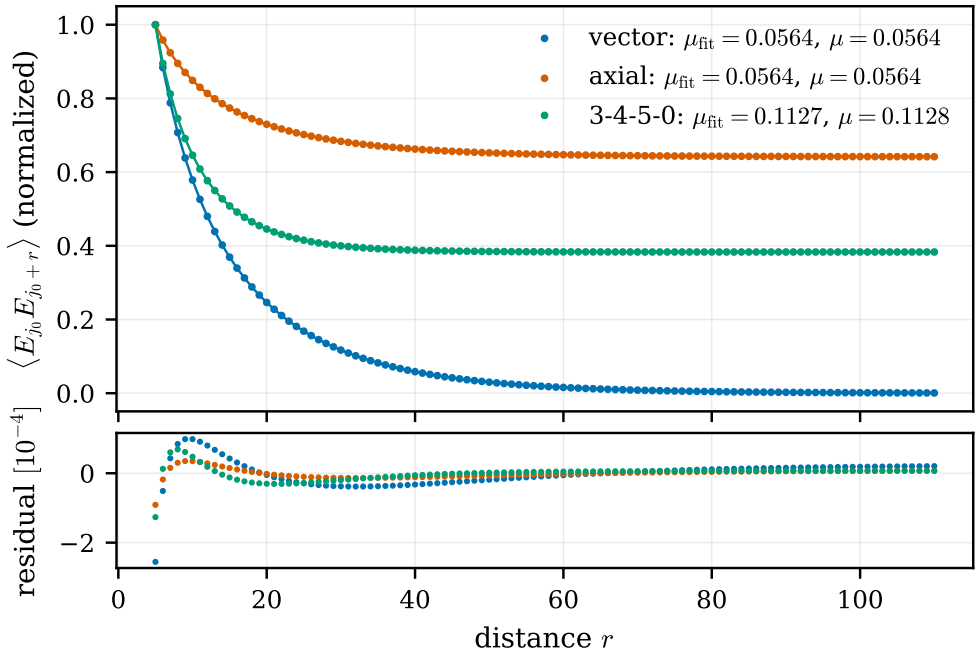


Figure 5.4: Equal-time correlator of the electric field in the bulk of the open chain ($N = 300$) with fits $AK_0(\mu r) + B$. The fitted masses agree with μ to about 10^{-3}.

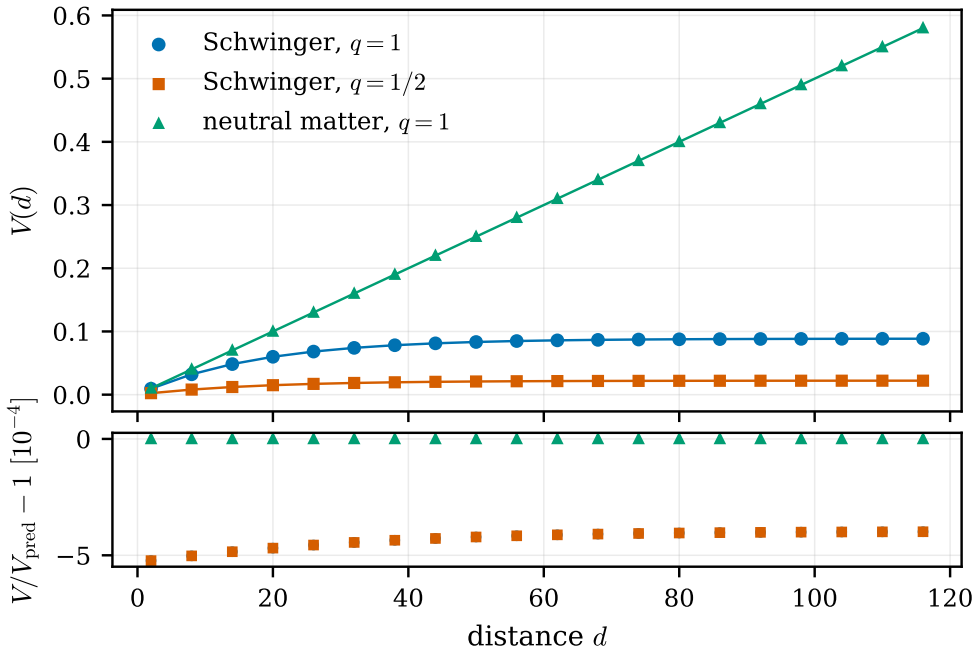


Figure 5.5: Static potential of probe charges $\pm q$ ($N = 400$, $g = 0.1$). Massless charged matter screens every charge, $V = (g^2q^2/2\mu)(1 - e^{-\mu d})$; without it the potential rises linearly (confinement), $V = g^2q^2d/2$.

5.5 Probe charges: screening and confinement

Place static external charges $+q$ at $x = 0$ and $-q$ at $x = d$. By the Gauss law they create the background $e(x) = gq$ for $0 < x < d$. The ground-state energy shift is the potential $V(d)$ between them. For the Schwinger model the Hamiltonian is $\int[\frac{1}{2}\Pi^2 + \frac{1}{2}(\partial\varphi)^2 + \frac{1}{2}(\mu\varphi + e)^2]$, and minimizing over φ in Fourier space gives

$$V(d) = \frac{1}{2}\int \frac{dk}{2\pi}\,|e_k|^2\,\frac{k^2}{k^2+\mu^2} = g^2q^2\int \frac{dk}{2\pi}\,\frac{1-\cos kd}{k^2+\mu^2} = \frac{g^2q^2}{2\mu}\,(1-e^{-\mu d}). \tag{5.9}$$

The potential saturates: the massless fermions *screen* any charge, even a fractional one [5]. With $\mu = g/\sqrt{\pi}$ the prefactor is $g^2q^2/2\mu = \frac{\sqrt{\pi}}{2}gq^2$, which is exactly the flat-space result quoted in [13], eq. (3.88); the lattice reproduces it to 5×10^{-4}. Without charged matter ($\mu = 0$) the same formula gives $V(d) = g^2q^2d/2$: a constant electric field between the charges, linear *confinement* with string tension $g^2q^2/2$. On the lattice the potential is the background energy $\min_{\boldsymbol{X}}[\frac{1}{2}\boldsymbol{X}^T M\boldsymbol{X} + \boldsymbol{b}^T\boldsymbol{X}] + \text{const} = \text{const} - \frac{1}{2}\boldsymbol{b}^T M^{+}\boldsymbol{b}$ (`api.screening_potential`); Fig. 5.5 shows both regimes.

5.6 The theta angle

A constant background field $e = g\theta/2\pi$ is the θ angle of the theory. With massless fermions it is unphysical: in the Schwinger model it is removed by the shift $\varphi \to \varphi - \sqrt{\pi}\,\theta/2\pi$, which corresponds to a chiral rotation of the fermions. On the ring this shift is a zero mode and costs nothing, so the background energy vanishes (numerically 10^{-15}). On the open chain the Dirichlet boundary fixes φ at the ends and a small boundary energy remains. A physical θ dependence requires massive fermions, which are beyond the Gaussian methods of this release.

5.7 In code

```
from latsimpkg import schwinger, ChiralTheory, spectrum, screening_potential

sp = spectrum(schwinger(g=0.04), N=300) # open chain
sp.max_deviation # ~1e-14: exact for vector charges
pot = screening_potential(schwinger(g=0.1), N=400, distances=range(2, 120, 6), q
    =0.5)
pot.meta["regime"], pot.max_rel_deviation # 'screening', ~5e-4
```

Chapter 6

Anomalies and global symmetries

Not every assignment of charges defines a consistent gauge theory. A chiral theory can suffer from a *gauge anomaly*: a symmetry of the classical theory that is necessarily broken by quantization. For a gauge symmetry this is fatal. On the lattice the anomaly shows up in a simple and concrete way: the electric field does not commute with itself.

6.1 The anomaly on the lattice

The Gauss law says that the gauge transformations are generated by $G(x) = \partial_x E - g\rho$. For a consistent theory these generators must commute; equivalently, the electric field determined by the Gauss law must satisfy $[E(x),\, E(y)] = 0$. Using $[\varphi_j,\, \vartheta_k] = i\mathbb{1}(j \le k)$, which follows from $\vartheta_k = \sum_{l\le k} P_l$, one finds for one gauge field

$$[E_j,\, E_k] = i\,\frac{g^2}{\pi}\,(Q^V \cdot Q^A)\,\operatorname{sign}(k-j), \qquad Q^V \cdot Q^A = \frac{1}{4}\Big(\sum Q_L^2 - \sum Q_R^2\Big). \tag{6.1}$$

The electric field commutes with itself if and only if

$$\sum_i Q_{L,i}^2 = \sum_j Q_{R,j}^2, \tag{6.2}$$

the anomaly-cancellation condition of two-dimensional chiral gauge theories [10]. The coefficient $\sum Q_L^2 - \sum Q_R^2$ is the anomaly: it counts the mismatch of the charged left- and right-movers. Figure 6.1 shows (6.1) for several anomalous charge assignments.

Example 6.1. The Schwinger model ($1^2 = 1^2$) and its axial version are anomaly free. The 3-4-5-0 model has $3^2 + 4^2 = 25 = 5^2 + 0^2$: a genuinely chiral but consistent theory. Charges $Q_L = (2, 1)$, $Q_R = (0, 1)$ give $5 - 1 = 4 \neq 0$ and are rejected by `ChiralTheory` unless `allow_anomalous=True`.

Several gauge fields. For $U(1)^k$ every pair of gauge fields must be anomaly free,

$$\mathcal{A}_{ab} = Q_{L,a} \cdot Q_{L,b} - Q_{R,a} \cdot Q_{R,b} = 0, \tag{6.3}$$

including the *mixed* anomalies $a \neq b$ (`ChiralTheory.anomaly_matrix`). In vector-axial form $\mathcal{A}_{ab} = 2(Q_a^V \cdot Q_b^A + Q_a^A \cdot Q_b^V)$.

A lattice subtlety for $k > 1$. On the lattice $[E_{a,j},\, E_{b,k}] = i\frac{g_a g_b}{\pi}\big(x_{ab}\mathbb{1}(j \le k) - x_{ba}\mathbb{1}(k \le j)\big)$ with $x_{ab} = Q_a^V \cdot Q_b^A$. Anomaly freedom only gives $x_{ab} + x_{ba} = 0$, so a constant commutator ix_{ab} between *different* gauge fields can remain. Since $x_{ab} = (Q_{R,a} \cdot Q_{L,b} - Q_{L,a} \cdot Q_{R,b})/4$ depends on how left and right movers are paired, it is an artefact of the regularization, absent for pairings with $Q_L Q_R^T$ symmetric. `build_gauge_model` searches for such a pairing automatically.

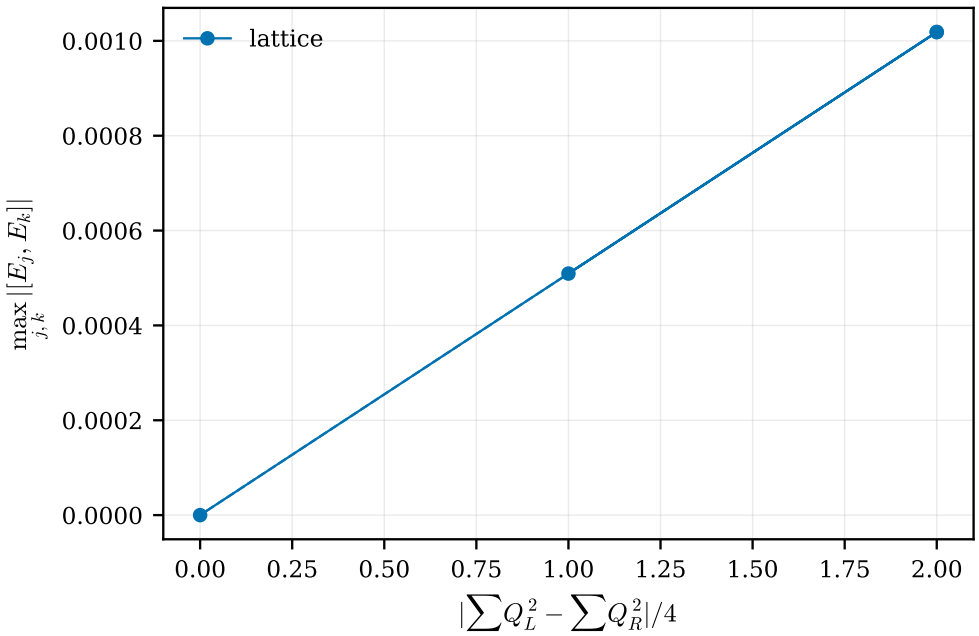


Figure 6.1: Commutator of the electric field for anomalous charge assignments against the anomaly coefficient $|\sum Q_L^2 - \sum Q_R^2|/4$. Dashed: g^2/π times the coefficient; it vanishes exactly for anomaly-free theories (3-4-5-0 at the origin).

6.2 Which global symmetries survive

Each fermion number is a global $U(1)$ symmetry of the free theory: $n_L + n_R$ of them, with charges $t = (t_L, t_R)$. After gauging, a global symmetry survives only if it has no *mixed* anomaly with the gauge field,

$$t_L \cdot Q_{L,a} = t_R \cdot Q_{R,a} \qquad \text{for all } a. \tag{6.4}$$

The others are broken by the anomaly, just as the axial charge of the Schwinger model is not conserved. For an anomaly-free theory the gauge charges themselves satisfy (6.4). The number of surviving symmetries is $n_L + n_R - \operatorname{rank}[Q_L, -Q_R]$ (`ChiralTheory.global_symmetries`): one for the Schwinger model (the gauge charge itself), three for the 3-4-5-0 model.

The lattice finds them by itself. On the ring the conserved charges are the kernel directions of MJ (Section 2.5). In terms of the fermion numbers, a conserved combination $t_n \cdot n + t_m \cdot m$ corresponds to $t_L \cdot N_L + t_R \cdot N_R$ with $t_n = (t_L + t_R)/2$, $t_m = (t_L - t_R)/2$, and (6.4) becomes $Q_a^A \cdot t_n + Q_a^V \cdot t_m = 0$. The tests verify that the conserved charges found numerically from the lattice Hamiltonian span exactly the space of (6.4): the anomaly structure is not put in by hand, it emerges from M.

Chapter 7

Charge sectors and the search for bound states

We now have all ingredients to answer the question of the introduction: which gauge-invariant particles does a theory contain? After the gauge field has given a mass to some combinations of the bosons, the remaining massless bosons are compact, and their charge sectors are the low-energy particles. Some of them are the original fermions, others are *composites*: gauge-invariant bound states of several fermions. The search is implemented in `find_bound_states`.

7.1 The idea

A state with fermion numbers $N_L = (N_{L,1}, \dots)$ and $N_R = (N_{R,1}, \dots)$ carries gauge charge

$$q_a = Q_{L,a} \cdot N_L + Q_{R,a} \cdot N_R = Q_a^V \cdot n + Q_a^A \cdot m, \tag{7.1}$$

with $n = N_L + N_R$ and $m = N_L - N_R$. On a ring the Gauss law $\oint \partial_x E = gq = 0$ allows only *neutral* sectors, $q_a = 0$ for all a. Neutral combinations of charged fermions are candidates for bound states. Whether such a sector really contains a light particle is decided by its energy: on a ring of length L a massless particle has energy $E \propto 1/L$, so

$$\Delta = \frac{EL}{2\pi} = h + \bar{h} \tag{7.2}$$

stays finite as $L \to \infty$ for a massless sector, while a sector that only contains massive excitations does not produce a $1/L$ level at all.

7.2 The lattice computation

On the ring every flavor has the zero modes $W_i = \sqrt{\pi} n_i$ and $P_{\mathrm{tot},i} = \sqrt{\pi} m_i$ (Section 3.4). The gauge field couples some combinations of them to the massive meson, so they are no longer conserved; the conserved charges are the kernel directions of MJ, i.e. the anomaly-free global symmetries of Chapter 6. The algorithm is:

1. Build the ring Hamiltonian M (`build_gauge_model` with `boundary="ring"`) and its conserved charges A (Section 2.5).
2. For every neutral (N_L, N_R) with $|N| \le n_{\max}$: the configuration $\boldsymbol{X}_v$ with $W_i = \sqrt{\pi} n_i$, $P_{i,j} = \sqrt{\pi} m_i / N$ has conserved charges $\boldsymbol{c} = A^T \boldsymbol{X}_v$.
3. Vectors with the same $\boldsymbol{c}$ are the same sector (they differ by excitations of the massive meson); keep the representative with the smallest $|N_L|^2 + |N_R|^2$.

4. The lowest energy in the sector is $E = \frac{1}{2}\boldsymbol{c}^T G \boldsymbol{c}$ (2.16): only the conserved charges are fixed, all other directions relax into the massive mode. The spin follows from the lattice momentum of the minimizing configuration, $s = -PL/2\pi$.

5. Classify: $\Delta = |s| = \frac{1}{2}$ is a single massless fermion; it is *elementary* if the representative is one of the original fermions and a *bound state* otherwise. All other sectors are multi-particle states.

Because the zero-mode energies are exact on the lattice (Chapter 4), Δ is independent of N and of $g > 0$: the result is not an extrapolation.

7.3 The analytic cross-check: the charge lattice

The same answer follows from a short argument. The gauge fields give a mass to the combinations $Q_{L,a} \cdot \varphi_L$ and $Q_{R,a} \cdot \varphi_R$ of the left- and right-moving bosons. What remains massless is the part of (N_L, N_R) orthogonal to the row spaces of Q_L and Q_R:

$$h = \tfrac{1}{2}|N_L^{\perp}|^2, \qquad \bar{h} = \tfrac{1}{2}|N_R^{\perp}|^2, \qquad s = h - \bar{h} = \tfrac{1}{2}(|N_L|^2 - |N_R|^2), \tag{7.3}$$

for neutral (N_L, N_R); the last equality uses neutrality. These vectors form the *charge lattice* of the low-energy theory (`analysis.charge_lattice`). `find_bound_states` computes both and reports whether they agree (`SectorScan.agrees`).

7.4 Worked example: the 3-4-5-0 model

The 3-4-5-0 model [11, 12] has two left movers with charges $3, 4$ and two right movers with charges $5, 0$ under one $U(1)$.

Anomaly
$3^2 + 4^2 = 25 = 5^2 + 0^2$: consistent, although no fermion is vector-like.

Pairing
$(3, 5)$ and $(4, 0)$: $Q^V = (4, 2)$, $Q^A = (-1, 2)$; indeed $Q^V \cdot Q^A = 0$.

Meson
$\mu^2 = \frac{g^2}{2\pi}(9 + 16 + 25 + 0) = \frac{25g^2}{\pi}$, one massive and one massless boson.

Global symmetries
four fermion numbers, one condition (6.4): three conserved charges, one of them the gauge charge.

The search over $|N| \le 3$, $\Delta \le 2$ finds twelve sectors, in exact agreement with the charge lattice (Fig. 7.1):

N_L	N_R	$(h, \bar{h})$	kind	content
$(2, 1)$	$(-2, 0)$	$(\frac{1}{2}, 0)$	bound state	$\psi_{3L}^2 \psi_{4L} \psi_{5R}^{-2}$
$(-2, -1)$	$(2, 0)$	$(\frac{1}{2}, 0)$	bound state	its antiparticle
$(0, 0)$	$(0, \pm 1)$	$(0, \frac{1}{2})$	elementary	$\psi_{0R}^{\pm 1}$
four sectors		$(\frac{1}{2}, \frac{1}{2})$	multi-particle	bound state $\times$ $\psi_{0R}^{\pm 1}$
four sectors		$(2, 0)$, $(0, 2)$	multi-particle	two identical fermions

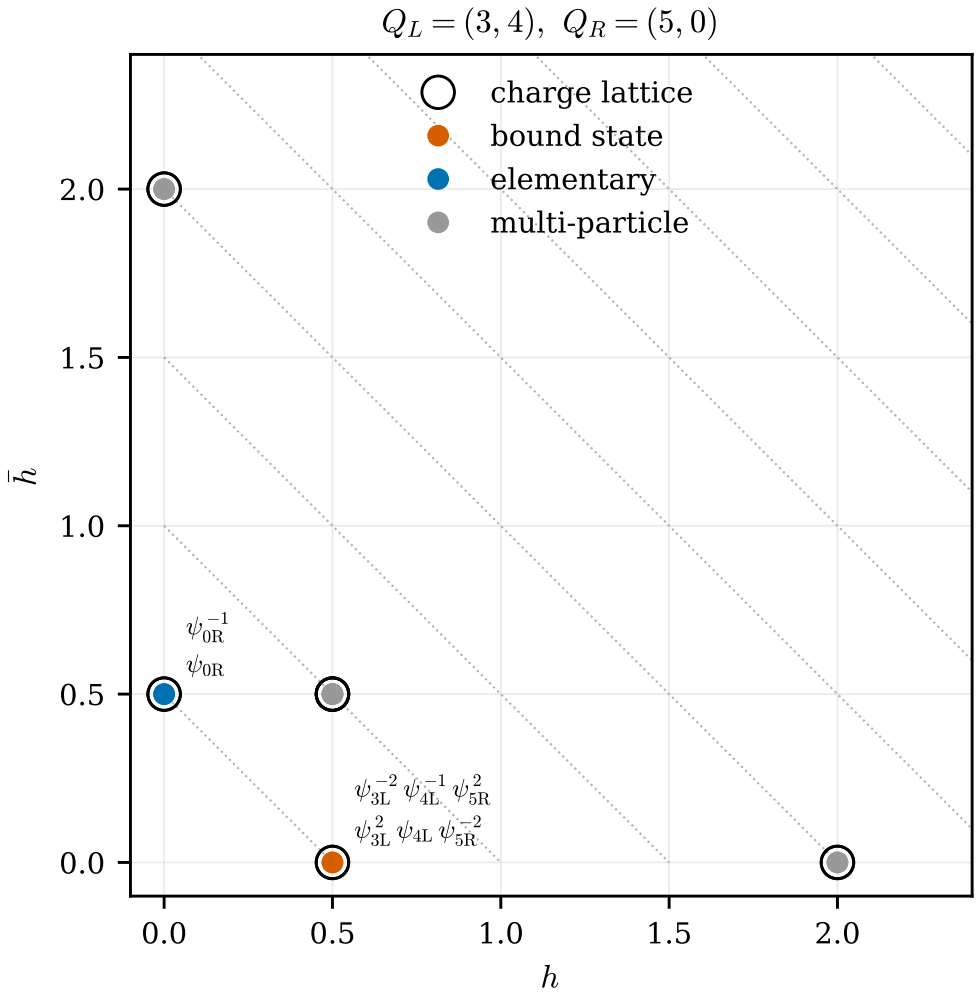


Figure 7.1: Gauge-invariant massless sectors of the 3-4-5-0 model in the $(h, \bar{h})$ plane. Filled: lattice (ring, $N = 60$); open circles: charge-lattice prediction. The bound states $\psi_{3L}^2 \psi_{4L} \psi_{5R}^{-2}$ and its antiparticle are left-moving fermions with $(h, \bar{h}) = (1/2, 0)$; ψ_{0R} is the elementary neutral right mover.

The bound state is neutral, $2 \cdot 3 + 1 \cdot 4 - 2 \cdot 5 = 0$, and left-moving with $h = \frac{1}{2}$: from (7.3), $N_L^\perp$ is the component of $(2, 1)$ orthogonal to $(3, 4)$, of length $|4 \cdot 2 - 3 \cdot 1|/5 = 1$, and $N_R^\perp = 0$. It is a massless gauge-invariant composite fermion. Together with the neutral right mover ψ_{0R} it forms the massless low-energy content of the theory: one left- and one right-moving fermion, central charges $(c_L, c_R) = (1, 1)$ instead of $(2, 2)$ before gauging. Counting particle and antiparticle, the theory has two bound states.

Comparison with the literature. The same four fermions carry a second anomaly-free $U(1)$, $H = (5, 0 \,|\, 3, -4)$, without mixed anomaly with $G = (3, 4 \,|\, 5, 0)$ [12]; the library finds H among the surviving global symmetries of the gauged 3-4-5-0 model. Tong gauges H and finds at low energies a single massless Dirac fermion, the elementary left mover $\Psi = \psi_2$ and the gauge-invariant composite right mover $\tilde{\Psi} = \psi_1^2 (\tilde{\psi}_1^\dagger)^2 \tilde{\psi}_2$, with G-charges $+4$ and -4 [12]. `find_bound_states(ChiralTheory([5,0],[3,-4]))` reproduces exactly this: ψ_2 is elementary with $(h, \bar{h}) = (\frac{1}{2}, 0)$, and the only bound states are $\psi_1^{\pm 2} \tilde{\psi}_1^{\mp 2} \tilde{\psi}_2^{\pm 1}$ with $(h, \bar{h}) = (0, \frac{1}{2})$ and G-charge ∓ 4. Gauging G instead, as in this chapter, is the same theory up to parity ($L \leftrightarrow R$) and charge conjugation of one fermion, and the sectors coincide with h and $\bar{h}$ exchanged (`tests/test_literature.py`). Wang and Wen [11] show that the 3-4-5-0 theory can be realized

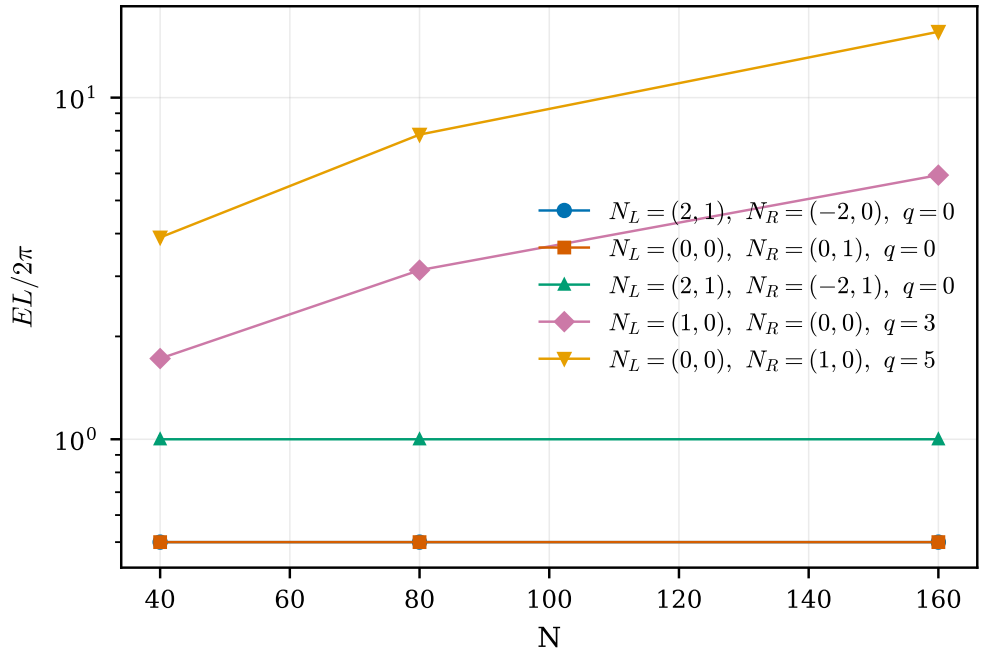


Figure 7.2: $EL/2\pi$ of charge sectors against the ring size (3-4-5-0, $g = 0.3$). Gauge-neutral massless sectors stay at $h + \bar{h}$; charged sectors (forbidden by the Gauss law on the ring) are screened, so E stays finite and EL grows.

as the low-energy limit of a local lattice model with an on-site $U(1)$ symmetry.

Volume dependence. Figure 7.2 shows $EL/2\pi$ against the ring size. The neutral massless sectors stay exactly at $h + \bar{h}$. Charged sectors are forbidden by the Gauss law on the ring; if one computes them anyway, the seam of the ring acts as an external counter-charge that the massless matter screens (Section 5.5), so E stays finite and EL grows: no massless particle.

7.5 Testing your own theory

```
from latsimpkg import ChiralTheory, find_bound_states
from latsimpkg.plotting import plot_sector_diagram, save_figure

theory = ChiralTheory(QL=[3, 4], QR=[5], g=0.3) # rejects anomalous charges
print(theory.anomaly_matrix, theory.meson_masses, theory.global_symmetries().shape
    [1])

scan = find_bound_states(theory, N=40, nmax=3, Delta_max=2.0)
assert scan.agrees # lattice = charge lattice
for s in scan.bound_states:
    print(s.content, s.h, s.hb)
save_figure(plot_sector_diagram(scan), "my_theory")
scan.save("my_theory.npz") # reload with SectorScan.load
```

In this example the right mover with charge 0 is missing; the pairing adds it as a neutral spectator (`spectator_R1`), and the bound states are those of the 3-4-5-0 model.
Limitations.

- Only massless fermions with abelian gauge fields: the Hamiltonian must be quadratic. Massive fermions or four-fermion interactions add cosine terms to the bosonized Hamiltonian

and require non-Gaussian methods.

- The search finds massless sectors. Massive bound states would be additional poles in correlation functions; in the Gaussian theories of this release the only massive particles are the free mesons.
- `nmax` limits which composites can be found: a bound state needs a representative with $|N_{L,i}|, |N_{R,i}| \leq n_{\max}$.
- The classification assumes a fermionic massless sector ($h \in k^2/2$), as for 3-4-5-0; for other theories inspect $(h, \bar{h})$ directly.

Appendix A

Conventions and checks

A.1 Conventions

All conventions are defined once in `latsimpkg.conventions`.

quantity	convention
units	$\hbar = c = 1$
phase space	$\boldsymbol{X} = (q_1..q_n, p_1..p_n)$, $[X_a,\, X_b] = iJ_{ab}$, $J = \begin{pmatrix} 0 & \mathbb{1} \\ -\mathbb{1} & 0 \end{pmatrix}$
Hamiltonian	$H = \frac{1}{2}\boldsymbol{X}^T M \boldsymbol{X}$ $(+\boldsymbol{b}^T\boldsymbol{X} + \text{const with background})$
lattice variables	$\varphi_j = \varphi(ja)$, $P_j = a\Pi(ja)$, $[\varphi_j,\, P_k] = i\delta_{jk}$
length	open chain $L = (N+1)a$, ring $L = Na$
dual field	$\vartheta_j = \sum_{k \le j} P_k$ $(+\ \bar{\vartheta}$ on the ring$)$
bosonization	$\rho_V = \partial_x\varphi/\sqrt{\pi}$, $\rho_A = \rho_L - \rho_R = \Pi/\sqrt{\pi}$, $\varphi \sim \varphi + \sqrt{\pi}$
fermion numbers	$n = N_L + N_R$ (winding), $m = N_L - N_R$ (momentum), $n \equiv m \bmod 2$
charges	$Q^V = (Q_L + Q_R)/2$, $Q^A = (Q_L - Q_R)/2$
Gauss law	$E_j - E_{j-1} = g\,\rho_j$ (+ probe charges), background $e = g\theta/2\pi$
spin	$s = -PL/2\pi = h - \bar{h}$, $P = -\frac{1}{a}\sum_j P_j \ell_j$; left movers have $s > 0$

A.2 What the test suite checks

The library is tested against analytic results (`pytest`; `pytest-mslow` for the expensive tests):

- linear algebra: oscillator moments, Williamson form on random symplectic matrices with known frequencies and free particles, symplecticity, purity of ground states, constrained minima, input validation;
- free field: Dirichlet spectrum to 10^{-12}, zero-point energy, ring sectors $\Delta = (N_L^2 + N_R^2)/2$ and $s = (N_L^2 - N_R^2)/2$;
- gauge theories: Schwinger spectrum exact, meson masses, a^2 continuum limit, anomalous commutator (6.1), correlator fit, screening and confinement potentials, θ on the ring;
- anomalies and symmetries: anomaly matrices, mixed anomalies, number of conserved charges, numerical conserved charges = analytic global symmetries;
- bound states: 3-4-5-0 sectors and bound states, lattice = charge lattice, independence of g and N, $U(1)^2$ examples;
- invariances: pairing, flavor permutation, charge conjugation;

- literature: the values of Appendix A.3;
- infrastructure: saving and loading of all results, plotting without global state, incremental figures.

A.3 Literature cross-check

Every reference value of the library was compared with the literature (`tests/test_literature.py`). All bibliographic data of the references were checked against the publishers or arXiv.

quantity	literature	source	lattice
Schwinger mass	$\mu = e/\sqrt{\pi}$	[3]	exact
static potential, massless Schwinger	$\frac{\sqrt{\pi}}{2} e q^2 (1 - e^{-\mu d})$	[13], eq. (3.88)	5×10^{-4}
screening of all charges	massless fermions screen	[5]	reproduced
string tension without matter	$V = e^2 q^2 d/2$	[13], eq. (3.2)	10^{-12}
anomaly cancellation	$\sum Q_L^2 = \sum Q_R^2$; $3^2 + 4^2 = 5^2$	[12, 11]	exact
second symmetry of 3-4-5-0	$H = (5, 0 \vert 3, -4)$, no mixed anomaly with G	[12], sec. 3	found
H gauged: IR Dirac fermion	$\Psi = \psi_2$, $\tilde{\Psi} = \psi_1^2 (\tilde{\psi}_1^\dagger)^2 \tilde{\psi}_2$, $G = \pm 4$	[12], sec. 3	exact
equal-time propagator	$\int \frac{dk}{2\pi} \frac{e^{ikx}}{2\omega_k} = \frac{K_0(\mu \lvert x \rvert)}{2\pi}$	numerical integration	10^{-6}

Appendix B

API reference

This chapter is generated from the docstrings of the library (`tools/build_docs.py`). The docstrings keep their formulas in plain ASCII so that they stay readable in an editor; the typeset mathematics behind them is in the preceding chapters.

B.1 Conventions: `latsimpkg.conventions`

Conventions of the bosonized theories, defined exactly once.

Bosonization (one Dirac fermion psi = (psi_L, psi_R) <-> one compact boson varphi):

```
rho_V = psi^dag psi       = d_x varphi / sqrt(pi)
rho_A = psi^dag g5 psi    = Pi / sqrt(pi)
varphi ~ varphi + sqrt(pi)   (compactness)
```

The factor sqrt(pi) is `SQRT_PI`.

Charges. A Dirac pair i combines the left mover L_i and the right mover R_i:

```
vector charge  Q^V = (Q_L + Q_R)/2,   axial charge  Q^A = (Q_L - Q_R)/2,
```

so that the gauge current Q_L j_L + Q_R j_R = Q^V j_V + Q^A j_A with j_V = j_L + j_R, j_A = j_L - j_R.

Fermion numbers. On the ring the fermion numbers are the zero modes of the boson:

```
winding   varphi(x + L) = varphi(x) + sqrt(pi) n,   n = integral of rho_V = N_L + N_R,
momentum  integral of Pi = sqrt(pi) m,              m = integral of rho_A = N_L - N_R.
```

N_L and N_R are integers, hence n = m (mod 2): (n, m) = (1, 0) would be "half a left and half a right fermion" and is not a state. This parity rule is what makes the compact boson fermionic.

Signs. Lattice momentum P = -(1/a) sum_j P_j l_j (continuum P = -int Pi d_x varphi) and spin s = -P L/(2 pi) = h - hb, chosen such that a single left mover (N_L, N_R) = (1, 0) has s = +1/2.

`vector_axial`

`vector_axial(QL, QR)`

Q^V = (Q_L + Q_R)/2, Q^A = (Q_L - Q_R)/2 (pairing L_i <-> R_i).

`chiral`

`chiral(QV, QA)`

Inverse of vector_axial: Q_L = Q^V + Q^A, Q_R = Q^V - Q^A.

winding_momentum

`winding_momentum(NL, NR)`

Fermion numbers -> winding n = N_L + N_R and zero-mode momentum m = N_L - N_R.

chiral_numbers

`chiral_numbers(n, m)`

Inverse: N_L = (n + m)/2, N_R = (n - m)/2. Raises ValueError unless n = m (mod 2).

B.2 Theories: `latsimpkg.theory`

The continuum input: chiral fermions with their U(1)^k charges, anomaly check, pairing into Dirac bosons, meson masses and surviving global symmetries.

class latsimpkg.theory.Pairing

dataclass with fields `QV`, `QA`, `pairs`, `spectators`

`Pairing(QV: numpy.ndarray, QA: numpy.ndarray, pairs: tuple, spectators: tuple) -> None`

Pairing of the chiral fermions into Dirac bosons.

Attributes

`QV (numpy.ndarray)`
Vector charges, shape (k, F).

`QA (numpy.ndarray)`
Axial charges, shape (k, F).

`pairs (tuple)`
For every Dirac boson the pair (L index, R index); None marks an added neutral spectator.

`spectators (tuple)`
Indices of the Dirac bosons that contain a spectator.

F *property*
Number of Dirac bosons (pairs, spectators included).

class latsimpkg.theory.ChiralTheory

`ChiralTheory(QL, QR, g=1.0, names=None, allow_anomalous=False, allow_fractional=False)`

A (1+1)-dimensional U(1)^k gauge theory of chiral fermions, given by their charges.

Charges: QL[a, i] is the charge of the left mover L_i under U(1)_a, QR[a, j] the one of the right mover R_j; g[a] is the coupling of U(1)_a.

Gauge anomaly: the two-point function of the gauge currents gives the k x k matrix

```
A_ab = Q_L,a . Q_L,b - Q_R,a . Q_R,b,
```

which must vanish for a consistent gauge theory. In vector-axial form (after pairing) A_ab = 2 (Q^V_a . Q^A_b + Q^A_a . Q^V_b), the coefficient of [E_a, E_b] on the lattice.

Meson masses: the electric energy (a/2) sum_a E_a^2 with E_a = (g_a/sqrt(pi)) (Q^V_a . varphi + Q^A_a . vartheta) gives at zero momentum the mass matrix

```
mu^2_ab = (g_a g_b / pi) (Q^V_a . Q^V_b + Q^A_a . Q^A_b) = (g_a g_b / 2pi) (Q_L,a . Q_L,b + Q_R,a
```

Global symmetries: a U(1) with charges t = (t_L, t_R) survives the gauging if it has no mixed anomaly with any gauge U(1), t_L . Q_L,a = t_R . Q_R,a for all a, i.e. t in the null space of [Q_L, -Q_R]. For an anomaly-free theory the gauge charges themselves are among them.

Arguments

`QL (array_like)`
Left charges, shape (k, n_L) or (n_L,) for k = 1.

`QR (array_like)`
Right charges, shape (k, n_R) or (n_R,) for k = 1.

`g (float | array_like)`
Couplings, one per U(1) (a scalar is used for all).

`names (tuple, optional)`
(names of the left movers, names of the right movers).

`allow_anomalous (bool, optional)`
Accept a gauge anomaly. Defaults to False.

`allow_fractional (bool, optional)`
Accept non-integer charges. Defaults to False.

Raises

`ValueError`
For inconsistent shapes, wrong number or negative couplings, fractional charges (unless allowed) or a gauge anomaly (unless allowed).

k *property*
Number of U(1) gauge fields.

n_L *property*
Number of left movers.

n_R *property*
Number of right movers.

anomaly_matrix *property*
A_ab = Q_L,a . Q_L,b - Q_R,a . Q_R,b.

is_anomaly_free *property*
True if the anomaly matrix vanishes (relative tolerance 1e-12).

pairing`(order=None)` *method*

Pair L_i with R_order[i] into Dirac bosons (default: in the given order).

If n_L != n_R the missing partners are added as neutral fermions (charge 0 under every U(1)). A neutral fermion is free and decoupled: it changes no anomaly and only adds an elementary sector.

Arguments

`order (sequence, optional)`
Permutation of the right movers.

Returns

Pairing: Q^V, Q^A of shape (k, F) with F = max(n_L, n_R), the pairs and the spectators.

meson_mass_matrix *property*

mu^2_ab = (g_a g_b / 2pi) (Q_L,a . Q_L,b + Q_R,a . Q_R,b).

meson_masses *property*

Sorted meson masses (square roots of the eigenvalues of mu^2).

global_symmetries() *method*

Basis of the U(1) symmetries without mixed anomaly with the gauge group: columns t = (t_L, t_R) (length n_L + n_R) with t_L . Q_L,a = t_R . Q_R,a for all a.

schwinger

schwinger(g=1.0)

Schwinger model: one Dirac fermion with vector charge 1 (Q_L = Q_R = 1).

axial_schwinger

axial_schwinger(g=1.0)

Axially gauged Schwinger model (Q_L = 1, Q_R = -1); equivalent to the vector one by charge conjugation of psi_R.

model_3450

model_3450(g=1.0)

3-4-5-0 model: left movers with charges 3, 4, right movers with 5, 0 (3^2 + 4^2 = 5^2 + 0^2).

B.3 Lattice: latsimpkg.lattice

Geometry of the spatial lattice: number of sites, spacing and boundary condition (open chain or ring).

class latsimpkg.lattice.Lattice

dataclass with fields ***N, a, boundary***

```
Lattice(N: int, a:  float = 1.0, boundary:  str = 'open') -> None
```

A one-dimensional spatial lattice of N sites with spacing a.

Boundary conditions:

```
"open": Dirichlet chain, varphi_{-1} = varphi_N = 0. The fixed end points lie outside
        the N sites, so the physical length is L = (N + 1) a and the free momenta are
        k_n = pi n / L (lattice: (2/a) sin(k_n a / 2)).
"ring": periodic chain, site N = site 0, so L = N a. The compact bosons keep their
        zero modes (windings and total momenta), i.e. the charge sectors.
```

Attributes

N (int)

Number of sites, N >= 2.

a (float)

Lattice spacing, a > 0.

`boundary (str)`
"open" or "ring".

`L` *property*
Physical length: (N + 1) a for the open chain, N a for the ring.

`is_ring` *property*
True for periodic boundary conditions.

B.4 Phase space: `latsimpkg.phasespace`

Layout of the phase-space vector X = (q, p) with named blocks, and the symplectic matrix J.

`class latsimpkg.phasespace.Block`

dataclass with fields ***name, conjugate, shape, offset***

`Block(name: str, conjugate: str, shape: tuple, offset: int) -> None`

A named block of canonical coordinates and its conjugate momenta.

Attributes

`name (str)`
Name of the coordinates, e.g. "phi".

`conjugate (str)`
Name of the conjugate momenta, e.g. "P".

`shape (tuple)`
Shape of the block, e.g. (F, N) for F flavors on N sites.

`offset (int)`
Position of the first coordinate in X.

`size` *property*
Number of coordinates in the block (product of the shape).

`class latsimpkg.phasespace.PhaseSpace`

`PhaseSpace(blocks)`

Layout of the phase-space vector X = (q, p) with named blocks.

All coordinates come first, then all momenta in the same order, so that [X_a, X_b] = i J_ab with the standard J = [[0, I], [-I, 0]]. Every coordinate block automatically has a conjugate momentum block at offset + n.

Example

```
ps = PhaseSpace([("phi", "P", (F, N)), ("theta_bar", "W", (F,))])
ps.index("phi", i, j)     # flat index of varphi_{i,j}
ps.slice("P", i)          # slice of P_{i,.}
ps.index("W", i)          # flat index of W_i
```

Attributes

`n (int)`
Number of degrees of freedom (coordinates).

`dim (int)`
Phase-space dimension 2n.

Arguments

`blocks (list)`
Tuples (name, conjugate, shape). Shapes are tuples of ints (an int is accepted for one-dimensional blocks).

Raises

`ValueError`
If a name is used twice or a shape is empty or non-positive.

`flat(n)` *classmethod*

Unstructured phase space with coordinates q_0..q_(n-1) and momenta p_0..p_(n-1).

`blocks` *property*
The blocks in phase-space order.

`index(name, *idx)` *method*

Flat index of one entry, e.g. index("phi", i, j).

Raises

`IndexError`
If the number of indices or their values do not fit the block.

`slice(name, *leading)` *method*

Contiguous slice of a block, fixing the leading indices, e.g. slice("phi", i) for varphi_{i,.} or slice("phi") for the whole block.

`zeros()` *method*

Zero phase-space vector.

`J` *property*
Symplectic matrix [[0, I], [-I, 0]], [X_a, X_b] = i J_ab.

B.5 Quadratic Hamiltonians: `latsimpkg.quadratic`

Quadratic Hamiltonians H = (1/2) X^T M X: normal modes, Williamson normal form including zero modes, Gaussian ground state, conserved charges and constrained minima. The numerical core of the library.

`class latsimpkg.quadratic.Tolerances`

dataclass with fields ***symmetry, psd, kernel, isotropy***

`Tolerances(symmetry: float = 1e-10, psd: float = 1e-10, kernel: float = 1e-10, isotropy: float = 1e-08) -> None`

All numerical thresholds in one place. Values are relative to the scale of M (its largest eigenvalue) unless stated otherwise.

Attributes

`symmetry (float)`
Allowed max|M - M^T| / max|M|.

`psd (float)`
Allowed negative eigenvalue of M, relative.

`kernel (float)`
Eigenvalues of M below this (relative) count as zero modes.

`isotropy (float)`
Allowed max|K^T J K| for the kernel K (absolute, K orthonormal).

class latsimpkg.quadratic.QuadraticHamiltonian

`QuadraticHamiltonian(M, phasespace=None, tol=Tolerances(symmetry=1e-10, psd=1e-10, kernel=1e-10, isotropy=1e-08))`

A quadratic Hamiltonian H = (1/2) X^T M X with [X_a, X_b] = i J_ab, M real symmetric and positive semidefinite. Immutable; expensive decompositions are computed once.

Decomposition (Williamson normal form including zero modes):

```
K = ker M (columns orthonormal) are the flat directions. They must be isotropic,
K^T J K = 0; then every zero mode is a free particle.
Equations of motion X' = J M X, so a linear charge w^T X is conserved iff M J w = 0,
i.e. the conserved charges are A = J K (orthonormal because J is orthogonal).
The oscillators live on W = K^omega cap K^perp = {X : K^T X = 0, K^T J X = 0}, with basis B.
There M_W = B^T M B is positive definite and Omega_W = B^T J B is the (non-standard)
symplectic form of the coordinates xi = B^T X. Every X splits uniquely as
    X = M^+ A G c + B xi + K kappa,   c = A^T X,   G = (A^T M^+ A)^(-1),
and the energy separates exactly,
    H = (1/2) c^T G c + (1/2) xi^T M_W xi.
The first term is the kinetic energy of the free particles (conserved momenta c),
the second a set of oscillators.
```

Williamson for (M_W, Omega_W): with L = M_W^(1/2) the matrix L Omega_W L is antisymmetric; the eigenvectors of the Hermitian i L Omega_W L give an orthogonal O and frequencies D with O^T (L Omega_W L) O = [[0, D], [-D, 0]]. Then S_W = L^(-1) O diag(D, D)^(1/2) satisfies

```
S_W^T M_W S_W = diag(D, D),   S_W J S_W^T = Omega_W,
```

so xi = S_W Y with canonical Y and H_W = sum_k omega_k (q_k^2 + p_k^2)/2.

Arguments

`M (array_like)`
The (2n x 2n) Hamiltonian matrix.

`phasespace (PhaseSpace, optional)`
Layout of X. Defaults to `PhaseSpace.flat(n)`.

`tol (Tolerances, optional)`
Numerical thresholds.

Raises

`ValueError`
If M is not square, has odd dimension, is not finite, not symmetric, not positive semidefinite, does not match the phase space, or has a non-isotropic kernel.

`J` *property*
Symplectic matrix of the phase space.

`kernel` *property*
Orthonormal basis K of ker M (flat directions), shape (2n, r).

`n_free` *property*
Number of free particles (zero modes) r.

`n_oscillators` *property*
Number of oscillators n - r.

`pinv` *property*
Moore-Penrose pseudo-inverse M^+.

`sqrt` *property*
M^(1/2).

`conserved_charges` *property*
Orthonormal basis A (columns) of the conserved linear charges A^T X: M J A = 0, A = J K.

`free_kinetic_matrix` *property*
G = (A^T M^+ A)^(-1): energy (1/2) c^T G c of the free particles with conserved charges c.

`free_energy(c)` *method*
Lowest energy (1/2) c^T G c in the sector with conserved charges c = A^T X.

`free_configuration(c)` *method*
Minimum-energy configuration X* = M^+ A G c with A^T X* = c (oscillators at rest).

`constrained_minimum(A, c)` *method*
Minimum of (1/2) X^T M X under the linear constraints A^T X = c (Lagrange multipliers):

```
M X = A lambda  =>  X* = M^+ A (A^T M^+ A)^(-1) c,  E = (1/2) c^T (A^T M^+ A)^(-1) c.
```

Arguments

`A (array_like)`
Constraint vectors as columns, shape (2n, m). Must be orthogonal to ker M.

`c (array_like)`
Constraint values, shape (m,).

Returns

tuple: (X*, E).

Raises

`ValueError`
If a constraint has a component along ker M (the minimum is then not attained in this form: the constraint can be satisfied at no cost along a flat direction).

`frequencies` *property*
Sorted oscillator frequencies omega_k > 0 (n - r values; free particles excluded).

`williamson` *property*
Symplectic normal form of the oscillators.

Returns

tuple: (omega, S) with the sorted frequencies omega (length n - r) and S of shape (2n, 2(n - r)) such that X_osc = S Y with canonical Y = (q_1.., p_1..), S^T M S = diag(omega, omega) and S J S^T = B Omega_W B^T.

ground_state_energy *property*

Zero-point energy (1/2) sum_k omega_k of the oscillators.

covariance *property*

Symmetrised ground-state covariance sigma_ab = (1/2) <{X_a, X_b}> = (1/2) S S^T.

For positive definite M this is the full ground state. With zero modes it is the covariance of the oscillator part in a sharp-momentum state of the free particles: it is exact for every observable w^T X with K^T w = 0, while the flat directions K themselves have infinite variance (they are not contained in sigma).

B.6 Gauge models: `latsimpkg.models.gauge`

Construction kit: ChiralTheory + Lattice -> quadratic lattice Hamiltonian of the bosonized gauge theory.

Fields (after pairing into F Dirac bosons): varphi_{i,j}, P_{i,j} = a Pi_i(x_j), vartheta_{i,j} = sum_{k<=j} P_{i,k}. Gauss law (temporal gauge), one electric field per U(1)_a:

```
E_{a,j} = (g_a/sqrt(pi)) sum_i [ Q^V_{a,i} varphi_{i,j} + Q^A_{a,i} vartheta_{i,j} ] + e_{a,j},
```

with a c-number background e_{a,j} = g_a theta_a/(2 pi) + g_a sum_{x<=j} q^ext_{a,x} (theta angle and static probe charges, E_j - E_(j-1) = g q^ext_j). Hamiltonian:

```
H = sum_{i,links} [ P^2 + l^2 ] / (2a) + (a/2) sum_{a,j} E_{a,j}^2 = (1/2) X^T M X + b^T X + cons
```

Boundaries:

```
"open": Dirichlet chain varphi_{-1} = varphi_N = 0, N + 1 links.
"ring": periodic, with the zero modes vartheta-bar_i and W_i = sqrt(pi) n_i (see ‘build_gauge_mod
```

Commutator of the electric fields: with [varphi_j, vartheta_k] = i 1(j <= k),

```
[E_{a,j}, E_{b,k}] = i (g_a g_b / pi) ( x_ab 1(j <= k) - x_ba 1(k <= j) ),   x_ab = Q^V_a . Q^A_b
```

Gauge-anomaly freedom gives x_ab + x_ba = A_ab / 2 = 0, so for a = b the commutator vanishes. For a != b a constant i x_ab remains; x_ab = (Q_R,a . Q_L,b - Q_L,a . Q_R,b)/4 depends on the pairing and is a regularization artefact. It vanishes for a pairing with Q_L Q_R^T symmetric, which `build_gauge_model` searches for.

`class latsimpkg.models.gauge.GaugeModel`

dataclass with fields ***theory, lattice, pairing, phasespace, H, electric_field, links, background, linear, constant***

```
GaugeModel(theory: latsimpkg.theory.ChiralTheory, lattice:
latsimpkg.lattice.Lattice, pairing: latsimpkg.theory.Pairing, phasespace:
latsimpkg.phasespace.PhaseSpace, H: latsimpkg.quadratic.QuadraticHamiltonian,
electric_field: numpy.ndarray, links: numpy.ndarray, background: numpy.ndarray,
linear: numpy.ndarray, constant: float) -> None
```

A built lattice model. All matrices act on the phase-space vector X of `phasespace`.

Attributes

`theory (ChiralTheory)`
The continuum input.

`lattice (Lattice)`
Geometry and boundary condition.

`pairing (Pairing)`
Dirac bosons used for the bosonization.

`phasespace (PhaseSpace)`
Layout of X.

`H (QuadraticHamiltonian)`
Quadratic part (1/2) X^T M X.

`electric_field (numpy.ndarray)`
C of shape (k, N, dim) with E_a = C_a X + e_a.

`links (numpy.ndarray)`
B of shape (F, n_links, dim) with the links l_i = B_i X.

`background (numpy.ndarray)`
e of shape (k, N) (theta angle and probe charges).

`linear (numpy.ndarray)`
b of the linear term b^T X = a sum_a e_a^T C_a X.

`constant (float)`
(a/2) sum e^2.

`M` *property*
The Hamiltonian matrix M of (1/2) X^T M X.

`F` *property*
Number of Dirac bosons (flavors).

`k` *property*
Number of U(1) gauge fields.

`QV` *property*
Vector charges of the pairing, shape (k, F).

`QA` *property*
Axial charges of the pairing, shape (k, F).

`describe()` *method*
Parameters of the model as plain dict (for the `meta` of results).

`gauge_charges(n, m)` *method*
Gauge charges q_a = Q^V_a . n + Q^A_a . m of the sector with windings n and momenta m.

`electric_field_commutator(a=0, b=0)` *method*
K with [E_{a,j}, E_{b,k}] = i K_jk.

`momentum(X)` *method*
Classical lattice momentum P = -(1/a) sum_i P_i^T l_i of the configuration X.

`mean_configuration` *property*
Shift <X> = -M^+ b of the ground state caused by the background (0 without background).

`background_energy` *property*

Classical energy of the background, min_X [(1/2) X^T M X + b^T X] + const = const - (1/2) b^T M^+ b.

ground_state_energy *property*

Zero-point energy of the oscillators plus the background energy.

choose_pairing

`choose_pairing(theory, order=None)`

Pairing for the lattice model. With `order` given it is used as is. Otherwise the given order is tried first, then all permutations of the right movers, until the electric fields of different U(1)s commute (x_ab = 0, see module docstring). For k = 1 every pairing works.

Raises

`ValueError`

If the theory is anomaly free but no pairing gives commuting electric fields.

build_gauge_model

`build_gauge_model(theory, lattice, theta=None, external_charges=None, pairing_order=None, tol=None)`

Build the lattice Hamiltonian of a bosonized U(1)^k gauge theory.

Ring: besides varphi~_{i,j}, P_{i,j} every flavor gets the dual zero mode vartheta-bar_i with conjugate W_i = sqrt(pi) n_i ([vartheta-bar_i, W_i] = i); varphi_{i,j} = varphi~_{i,j} + W_i j/N and vartheta_{i,j} = vartheta-bar_i + sum_{k<=j} P_{i,k}, giving [varphi_j, vartheta_k] = i (1(j <= k) - j/N). The winding and the total momentum are the fermion numbers (see `conventions`).

Arguments

`theory (ChiralTheory)`

Charges and couplings.

`lattice (Lattice)`

N, a and the boundary condition.

`theta (float | array_like, optional)`

Theta angle(s), constant background field g theta/(2 pi).

`external_charges (array_like, optional)`

Static probe charges per site, shape (k, N) or (N,) for k = 1.

`pairing_order (sequence, optional)`

Fixed pairing of the right movers (default: automatic).

`tol (Tolerances, optional)`

Passed to `QuadraticHamiltonian`.

Returns

GaugeModel: The built model.

B.7 Result types: `latsimpkg.analysis.results`

Result types of the analysis layer.

Every result is an immutable dataclass that carries its data and the parameters it was computed with (`meta`), and can be saved and loaded without recomputation:

```
result.save("spectrum.npz");  Spectrum.load("spectrum.npz")
```

Arrays are stored as arrays in the .npz file, everything else as JSON under the key "___meta___".

`to_jsonable`

`to_jsonable(value)`

Recursively convert numpy scalars/arrays and tuples into plain JSON types.

`class latsimpkg.analysis.results.Sector`

dataclass with fields `NL`, `NR`, `Delta`, `s`, `kind`, `content`

```
Sector(NL: tuple, NR: tuple, Delta:  float, s:  float, kind:  str, content:  str = ")
-> None
```

A gauge-invariant charge sector, labelled by its lightest representative (N_L, N_R).

Attributes

`NL, NR (tuple)`
Chiral fermion numbers per Dirac boson of the pairing.

`Delta (float)`
Scaling dimension h + hb = E L / (2 pi).

`s (float)`
Spin h - hb.

`kind (str)`
"elementary", "bound state" or "multi-particle" (see `classify_sector`).

`content (str)`
The sector as product of fermion operators, e.g. "psi_3L^2 psi_4L psibar_5R^2".

`h` *property*
Left-moving conformal weight h = (Delta + s)/2.

`hb` *property*
Right-moving conformal weight hb = (Delta - s)/2.

`key` *property*
(NL, NR): the representative, used to match numerics and analytics.

`as_dict()` *method*
Plain dict including h and hb.

`to_json()` *method*
JSON-serializable dict (used by `SectorScan.save`).

`from_json(d)` *classmethod*
Inverse of `to_json`.

sector_signature

`sector_signature(sectors, digits=6)`

Representation-independent fingerprint of a sector list: sorted (Delta, s, kind).

class latsimpkg.analysis.results.SectorEnergy

dataclass with fields `q, E, Delta, s`

`SectorEnergy(q: numpy.ndarray, E: float, Delta: float, s: float) -> None`

Lowest energy in the sector (N_L, N_R) of a ring model.

Attributes

`q (numpy.ndarray)`
Gauge charges, shape (k,); a physical sector on the ring needs q = 0.

`E (float)`
Energy above the vacuum.

`Delta (float)`
E L / (2 pi).

`s (float)`
Spin from the lattice momentum, -P L / (2 pi).

class latsimpkg.analysis.results.SectorScan

dataclass with fields `numeric, analytic, meta`

`SectorScan(numeric: tuple, analytic: tuple, meta: dict = <factory>) -> None`

Result of a bound-state search: numerical sectors on the ring and the analytic charge-lattice prediction.

Attributes

`numeric (tuple[Sector])`
Sectors found from the lattice Hamiltonian.

`analytic (tuple[Sector])`
Sectors predicted by the charge-lattice analysis.

`meta (dict)`
Theory, lattice and scan parameters.

bound_states *property*
Numerical sectors classified as bound states.

elementary *property*
Numerical sectors that are single elementary fermions.

multi_particle *property*
Numerical sectors that are multi-particle states.

agrees *property*
Numerics and analytics give the same sectors (same Delta, s and kind).

max_deviation *property*
Largest |Delta_num - Delta_ana| and |s_num - s_ana| over sectors with the same representative.

class latsimpkg.analysis.results.Spectrum

dataclass with fields ***frequencies, prediction, meta***

`Spectrum(frequencies: numpy.ndarray, prediction: numpy.ndarray, meta: dict = <factory>) -> None`

Normal-mode frequencies of a model.

Attributes

`frequencies (numpy.ndarray)`
Sorted oscillator frequencies.

`prediction (numpy.ndarray)`
Continuum-motivated prediction of the same length (empty if none).

`meta (dict)`
Theory and lattice parameters, number of free particles.

`omega_min` *property*
Lowest frequency.

`max_deviation` *property*
max |omega - prediction| (nan without prediction).

`count_below(value)` *method*
Number of frequencies below `value`.

class latsimpkg.analysis.results.Series

dataclass with fields ***x, y, prediction, meta***

`Series(x: numpy.ndarray, y: numpy.ndarray, prediction: numpy.ndarray, meta: dict = <factory>) -> None`

A measured quantity y(x) with an optional prediction, e.g. the lowest frequency against N.

Attributes

`x, y (numpy.ndarray)`
Data.

`prediction (numpy.ndarray)`
Prediction at x (empty if none).

`meta (dict)`
Labels ("name", "xlabel", "ylabel") and parameters.

`max_deviation` *property*
max |y - prediction| (nan without prediction).

class latsimpkg.analysis.results.FitWindow

dataclass with fields ***j0, r_min, r_max***

`FitWindow(j0: int, r_min: int, r_max: int) -> None`

Window of a correlator fit: reference site j0 and distances r_min <= r < r_max.

`FitWindow.bulk(N)` keeps j0 = N/2 - N/5 and r < 0.37 N, i.e. all sites at least ˜0.2 N away from the open boundaries.

`bulk(N)` *classmethod*
Default window for N sites (see class docstring).

`check(N)` *method*
Raise ValueError if the window does not fit into N sites.

class latsimpkg.analysis.results.FitResult

dataclass with fields ***mu, A, B, mu_expected, r, y, meta***

```
FitResult(mu: float, A: float, B: float, mu_expected: float, r: numpy.ndarray, y:
numpy.ndarray, meta: dict = <factory>) -> None
```

Fit of the electric-field correlator by A K_0(mu r) + B.

Attributes

`mu (float)`
Fitted meson mass.

`A, B (float)`
Amplitude and constant offset.

`mu_expected (float)`
Continuum meson mass.

`r (numpy.ndarray)`
Distances (physical units) used in the fit.

`y (numpy.ndarray)`
Correlator values at r.

`meta (dict)`
Fit window, theory and lattice parameters.

rel_deviation *property*
|mu - mu_expected| / mu_expected.

model(r) *method*
Fitted curve A K_0(mu r) + B.

class latsimpkg.analysis.results.Potential

dataclass with fields ***d, V, prediction, meta***

```
Potential(d: numpy.ndarray, V: numpy.ndarray, prediction: numpy.ndarray, meta:
dict = <factory>) -> None
```

Static potential V(d) between probe charges +q and -q at distance d.

Attributes

`d (numpy.ndarray)`
Distances (physical units).

`V (numpy.ndarray)`
Energy of the background (ground-state energy shift).

`prediction (numpy.ndarray)`
Continuum prediction (empty if none).

`meta (dict)`
q, theory and lattice parameters, "regime" ("screening" or "confinement").

max_rel_deviation *property*
max |V / prediction - 1| (nan without prediction).

B.8 Spectra: `latsimpkg.analysis.spectrum`

Normal-mode spectra, their continuum prediction and scaling with the lattice size.

`dirichlet_momenta`

`dirichlet_momenta(N, a=1.0)`

Free frequencies of the Dirichlet chain, (2/a) sin(pi n / (2(N+1))), n = 1..N (k_n = pi n / L).

`predicted_spectrum`

`predicted_spectrum(model)`

Continuum-motivated spectrum on the open chain: for every meson mass mu_a a branch sqrt(mu_a^2 + k^2), and F - (number of mesons) massless branches k, with the Dirichlet lattice momenta k.

Exact for vector-like charges (they act locally on varphi). Axial charges act on vartheta = cumulative P, whose boundary conditions differ: their momenta are k_n = (n + delta) pi / L with a phase shift delta, so the prediction is only correct in the bulk (deviation ~ 1/N). Returns an empty array on the ring.

`mode_spectrum`

`mode_spectrum(model)`

Normal-mode frequencies of `model` with the prediction of `predicted_spectrum`.

`lowest_frequency_scaling`

`lowest_frequency_scaling(theory, Ns, a=1.0)`

omega_min against N on the open chain.

Prediction: with a meson branch only (F = number of mesons) omega_min = sqrt(mu^2 + k_1^2) -> mu, otherwise the massless branch gives omega_min = k_1 = (2/a) sin(pi/(2(N+1))) -> 0.

`continuum_scaling`

`continuum_scaling(theory, L, spacings)`

omega_min at fixed physical length L = (N + 1) a for decreasing a (open chain).

Prediction: sqrt(mu^2 + (pi/L)^2) for the lightest meson branch; the lattice dispersion (2/a) sin(k a / 2) deviates by O((k a)^2), so the deviation falls like a^2.

B.9 Correlators: `latsimpkg.analysis.correlators`

Correlation functions of the electric field.

`electric_field_correlator`

`electric_field_correlator(model, a=0, b=0)`

Ground-state correlator <E_{a,j} E_{b,k}> (connected, symmetrised) = (C_a sigma C_b^T)_jk.

meson_mass_from_E

`meson_mass_from_E(model, window=None, mu_guess=None, u1=0)`

Fit the ground-state correlator <E_{j0} E_{j0+r}> by A K_0(mu r a) + B.

Physics: the electric field is proportional to the massive boson, and the equal-time propagator of a free massive boson in 1+1 dimensions is K_0(mu |x|) / (2 pi), so the decay of <E E> measures mu. The offset B is a uniform fluctuation of the background field caused by the open boundary (non-zero for axial charges).

Arguments

`model (GaugeModel)`
Model on the open chain.

`window (FitWindow, optional)`
Reference site and distances. Defaults to `FitWindow.bulk(N)`.

`mu_guess (float, optional)`
Start value. Defaults to the lightest continuum meson mass.

`u1 (int, optional)`
Which U(1). Defaults to 0.

Returns

FitResult

B.10 Potentials: `latsimpkg.analysis.potentials`

Static potentials between probe charges: screening and confinement.

potential_prediction

`potential_prediction(theory, q, d, u1=0)`

Continuum potential of probe charges +q, -q at distance d (one U(1)):

```
mu > 0:  V(d) = (g^2 q^2 / 2 mu) (1 - exp(-mu d))   (screening by the massless fermions)
mu = 0:  V(d) = g^2 q^2 d / 2                        (confinement, string tension g^2 q^2 / 2)
```

Derivation: minimizing (1/2) int [(d varphi)^2 + (mu varphi + e)^2] for the probe field e = g q 1_[0,d] gives V = (g^2 q^2) int dk/2pi (1 - cos k d) / (k^2 + mu^2). Returns None for k > 1 (no closed formula implemented).

screening_potential

`screening_potential(theory, N, distances, q=1.0, a=1.0, j0=None, u1=0)`

V(d) on the open chain: probe charges +q at site j0 and -q at site j0 + d, energy of the background (ground-state energy shift, see `GaugeModel.background_energy`).

Arguments

`theory (ChiralTheory)`
The theory.

`N (int)`
Number of sites; keep j0 and j0 + d far (» 1/mu) from the boundaries.

`distances (sequence[int])`
Distances in lattice units.

`q (float, optional)`
Probe charge (fractional values allowed). Defaults to 1.

`a (float, optional)`
Lattice spacing. Defaults to 1.

`j0 (int, optional)`
Site of +q. Defaults to N // 2 - max(distances) // 2.

`u1 (int, optional)`
Which U(1) the probes are charged under. Defaults to 0.

Returns

Potential

B.11 Charge sectors: `latsimpkg.analysis.sectors`

Charge sectors of a gauge model on the ring.

The fermion numbers of Dirac boson i are its zero modes: winding W_i = sqrt(pi) n_i and total momentum sum_j P_{i,j} = sqrt(pi) m_i, with N_L = (n + m)/2, N_R = (n - m)/2 (see `conventions`). Only the conserved combinations c = Aˆ T X_v are good quantum numbers (A from the kernel of M, `QuadraticHamiltonian`); the lowest energy in a sector is the free-particle energy (1/2) cˆ T G c of the Williamson normal form. Massless sectors have E L/(2 pi) = h + hb independent of L, the momentum gives the spin s = h - hb.

`sector_state`

`sector_state(model, n, m)`

Configuration X_v of the sector (n, m): W_i = sqrt(pi) n_i, P_{i,j} = sqrt(pi) m_i / N.

`charge_map`

`charge_map(model)`

Matrix T with c = T (n, m): the values of the conserved charges Aˆ T X_v in the sector (n, m). Its rows (t_n, t_m) are the conserved combinations of the fermion numbers; for an anomaly-free gauge theory they satisfy Qˆ A_a . t_n + Qˆ V_a . t_m = 0 (no mixed anomaly with any gauge U(1)).

`sector_energy`

`sector_energy(model, NL, NR)`

Energy of the lowest state in the sector with chiral fermion numbers (N_L, N_R).

Returns

SectorEnergy: gauge charges q (shape (k,)), E, Delta = E L/(2 pi), s = -P L/(2 pi).

sector_scan

`sector_scan(model, nmax=3, Delta_max=2.0)`

Search all gauge-neutral charge sectors with |N_L|, |N_R| <= nmax per component.

Vectors v with the same conserved charges c describe the same sector; the one with the smallest |N_L|^2 + |N_R|^2 is kept as representative (see `classify_sector`).

Arguments

`model (GaugeModel)`
A model on the ring.

`nmax (int, optional)`
Range of the fermion numbers. Defaults to 3.

`Delta_max (float, optional)`
Keep sectors with E L/(2 pi) <= Delta_max. Defaults to 2.0.

Returns

list[Sector]: Sorted by Delta, then by decreasing s.

B.12 Charge lattice: `latsimpkg.analysis.charge_lattice`

Analytic charge-lattice analysis of the gauge-invariant massless sectors (continuum prediction).

fermion_content

`fermion_content(NL, NR, theory=None, pairing=None)`

Write a sector as product of fermion operators, e.g. "psi_3L^2 psi_4L psi_5R^-2".

The exponent is the fermion number (negative: antiparticles). With `theory` and `pairing` the names of the theory are used; added spectators are called "spectator_L<i>" / "spectator_R<i>".

classify_sector

`classify_sector(NL, NR, Delta, s, content='', tol=1e-06)`

Collect the data of a sector and classify it.

A sector with h + hb = 1/2 and |s| = 1/2 is a single massless chiral fermion. It is "elementary" if its lightest representative is one of the original fermions (|N_L|^2 + |N_R|^2 = 1), otherwise a "bound state" (gauge-invariant composite). All other sectors are "multi-particle" states built from these fermions. (This criterion assumes that the massless sector is fermionic, h in k^2/2, as for 3-4-5-0.)

Returns

Sector

charge_lattice_spectrum

`charge_lattice_spectrum(pairing, nmax=3, Delta_max=2.0, theory=None)`

Continuum prediction for the gauge-invariant massless sectors.

With the chiral charges Q_L = Q^V + Q^A, Q_R = Q^V - Q^A (rows a = 1..k) a state with fermion numbers (N_L, N_R) is gauge neutral if Q_L,a . N_L + Q_R,a . N_R = 0 for all a. The gauge fields give a mass to the combinations Q_L,a . varphi_L and Q_R,a . varphi_R, so only the parts orthogonal to the row spaces of Q_L and Q_R stay massless:

```
h = |N_L^perp|^2 / 2,   hb = |N_R^perp|^2 / 2,   s = h - hb = (|N_L|^2 - |N_R|^2) / 2.
```

Vectors with the same (N_L^perp, N_R^perp) describe the same sector. The fermion numbers refer to the Dirac bosons of `pairing` (spectators included), exactly as in the numerical sector scan.

Arguments

`pairing (Pairing)`
Charges of the Dirac bosons, e.g. `theory.pairing()` or `model.pairing`.

`nmax (int, optional)`
Range of the fermion numbers. Defaults to 3.

`Delta_max (float, optional)`
Keep sectors with h + hb <= Delta_max. Defaults to 2.0.

`theory (ChiralTheory, optional)`
Only used to name the fermions in `Sector.content`.

Returns

list[Sector]: Sorted by Delta, then by decreasing s.

`sort_sectors`

`sort_sectors(sectors)`

Canonical order: increasing Delta, decreasing s, then by representative.

B.13 User API: `latsimpkg.api`

User-facing functions: one call from a theory to a result.

```
from latsimpkg import model_3450, find_bound_states
scan = find_bound_states(model_3450(g=0.3))
for s in scan.bound_states:
    print(s.content, s.h, s.hb)
```

`find_bound_states`

`find_bound_states(theory, N=40, a=1.0, nmax=3, Delta_max=2.0)`

Search the gauge-invariant massless sectors of a theory and identify its bound states.

Builds the theory on a ring of N sites (zero modes = fermion numbers), scans all gauge-neutral fermion numbers |N_L|, |N_R| <= nmax, groups them by their conserved charges, reads off Delta = h + hb and the spin s, and classifies every sector as elementary fermion, bound state (gauge-invariant composite fermion) or multi-particle state. The analytic charge-lattice prediction is computed alongside as an independent cross-check (`SectorScan.agrees`).

The massless sectors do not depend on g > 0 or on N (E L/(2 pi) is exact on the lattice), so small N suffices; nmax limits which composites can be found.

Arguments

`theory (ChiralTheory)`
The theory (must be anomaly free).

`N (int, optional)`
Ring size. Defaults to 40.

`a (float, optional)`
Lattice spacing. Defaults to 1.

`nmax (int, optional)`
Range of the fermion numbers. Defaults to 3.

`Delta_max (float, optional)`
Largest Delta kept. Defaults to 2.

Returns

SectorScan: numerical and analytic sectors; `.bound_states`, `.elementary`, `.agrees`.

`spectrum`

`spectrum(theory, N, a=1.0, boundary='open', **build_kwargs)`

Normal-mode spectrum of a theory on a lattice, with the continuum-motivated prediction (open chain; see `analysis.spectrum.predicted_spectrum`).

Returns

Spectrum

`screening_potential`

`screening_potential(theory, N, distances, q=1.0, a=1.0, j0=None, u1=0)`

Static potential V(d) of probe charges +q, -q (open chain) with the continuum prediction: screening (1 - exp(-mu d)) with charged massless matter, linear confinement without.

Returns

Potential

B.14 Plotting: `latsimpkg.plotting`

Plots of saved results.

Every function takes result objects (`Spectrum`, `Series`, `FitResult`, `Potential`, `SectorScan`) and returns a matplotlib `Figure`; nothing is recomputed. The figures are built with the object-oriented API (no pyplot, no global state, no backend switching), so plotting is fast, works headless and does not interfere with interactive sessions.

```
fig = plot_sector_diagram(SectorScan.load("results/sectors/scan_3_4_5_0.npz"))
save_figure(fig, "figures/sectors_3450")          # writes .pdf (vector, for LaTeX) and .png
```

`new_figure`

`new_figure(residuals=False, size=(5.0, 3.3))`

Figure with one axis, or a main axis plus a smaller residual axis below it.

`save_figure`

`save_figure(fig, stem, formats=('pdf', 'png'), dpi=200)`

Write `stem`.pdf / `stem`.png (directories are created). Returns the written paths.

content_to_tex

`content_to_tex(content)`

'psi_3L^2 psi_4L psi_5R^-2' -> mathtext '$\psi_{3L}^{2}\,\psi_{4L}\,\psi_{5R}^{-2}$'.

theory_label

`theory_label(meta)`

Short label of a theory from result meta, e.g. 'Q_L = (3, 4), Q_R = (5, 0)'.

plot_spectrum

`plot_spectrum(spectra, labels=None, residuals=True)`

Normal-mode frequencies omega_n against the mode index n, with the prediction (dashed).

plot_series

`plot_series(series, labels=None, residuals=True, logx=False, logy=False, abs_residuals=False, xlabel=None, ylabel=None)`

y(x) with the prediction (dashed); optional residual panel (|y - prediction| on a log scale if abs_residuals).

plot_fits

`plot_fits(fits, labels=None, residuals=True)`

Electric-field correlators (normalized to the first point) with their fits A K_0(mu r) + B.

plot_potentials

`plot_potentials(potentials, labels=None, residuals=True)`

Static potentials V(d) of probe charges with the continuum prediction; residual panel V/V_pred - 1.

plot_sector_diagram

`plot_sector_diagram(scan, annotate=('bound state', 'elementary'), size=(4.6, 4.2))`

Massless sectors in the (h, hb) plane: filled points = numerics (coloured by kind), open circles = charge-lattice prediction, dotted lines = constant Delta = h + hb. Sectors of the given kinds are labelled with their fermion content.

plot_result

`plot_result(result, **kwargs)`

Default plot for any result type.

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